1 Introduction

Racial and socioeconomic inequality in college access is substantially rooted in the racial and socioeconomic stratification of residential space (Domina et al. 2017; Stevens et al. 2008; Reardon and Bischoff 2011; Korver-Glenn 2022). On the demand side, where students live affects their educational resources and the peer and parental networks that influence college-going behavior (Sewell and Shah 1967; Bourdieu 1984; Huang 2023; McDonough 1997). On the supply side, stratified geography also shapes where colleges recruit and which prospective students they contact. However, translating stratified geography into college behavior requires infrastructures that classify places and people and define student markets. Third-party intermediaries build and circulate these infrastructures, connecting stratified geography to organizational action.

Scholarship on college access has paid little attention to these mediating infrastructures. Recent syntheses show that the literature centers familiar actors—students and families, schools, colleges, and policymakers—and institutional processes such as advising, recruiting, admissions, financial aid, and college choice (Domina et al. 2017; Stevens et al. 2008). Even studies showing that colleges disproportionately recruit from affluent, predominantly White schools and communities treats recruiting as behavior undertaken by individual colleges (Stevens 2007; Author XXXXa, XXXXb; Khan 2011). Taken together, this literature moves from stratified geography to college behavior without considering the intermediating vendors, products, and infrastructures that classify places and people on behalf of colleges looking for customers.

College Board exemplifies this intermediary role. Working with a consortium of elite private colleges, it underwrote the development of Geomarkets and the Market Segment Model, formalizing admissions officers' shared understanding of stratified student markets (Zemsky and Oedel 1983). Geomarkets divide states and metropolitan areas into local recruiting territories, while the Market Segment Model provides a class-centered account of student

demand across those territories. College Board commercialized these classifications through its recruiting products, making them available to colleges beyond the originating network. For example, its Enrollment Planning Service (EPS) was software-as-a-service (SAAS) recruiting product based on the Market Segment Model, helped colleges identify which Geomarkets to target and which high schools to visit within them (Takamiya 2005; College Board 2019). Geomarkets remain embedded in contemporary college recruiting, including the customer relationship management systems colleges use to assign admissions staff to recruiting territories (Technolutions 2026a; Oracle n.d.).

This manuscript focuses on Geomarkets' integration into Student Search Service, College Board's product for selling "student lists" containing prospective students' contact information. Student lists are a fundamental input for undergraduate recruiting campaigns because purchased names – alongside prospects who contact colleges on their own – constitute the population available for subsequent recruiting interventions (Ruffalo Noel-Levitz 2022b). Among surveyed bachelor's-granting institutions, 86% of public and 87% of private nonprofit colleges reported purchasing student lists (Ruffalo Noel-Levitz 2022b). Colleges determine which prospect profiles to purchase by applying search filters such as graduation year, test score, and geography. By incorporating Geomarkets as a filter, College Board embedded its classification of stratified space within the process through which colleges select prospective students to contact (College Board 2001).

Drawing on the market devices framework (Muniesa et al. 2007; Hirschman and Garbes 2019), we argue that market devices developed by College Board mediate between stratified geography and organizational action by colleges looking for customers. The "affordances" of a market device are the actions it makes possible. Each affordance is associated with an outcome. The consequence of choosing one affordance over another is the difference between the outcomes associated with those choices. Student Search Service is a selection device whose filters help constitute the prospect populations available for colleges' recruiting efforts

(Hirschman and Bosk 2020; Author XXXXc). The Geomarket filter turns College Board's proprietary recruiting territories into criteria for including or excluding individual prospective students. Given racial and socioeconomic residential stratification, this filtering affordance can make different groups of academically similar students more or less visible to colleges.

Yet the consequences of Geomarket filtering are not fully visible within Student Search Service. When colleges purchase a student list, they observe only the profiles generated by the filters they select, not the counterfactual prospect pools associated with alternative filter selections (Author XXXXd). We reconstruct these counterfactual prospect pools to investigate how Geomarket filtering can shape the racial/ethnic and first-generation composition of academically similar prospect pools. Holding academic criteria constant, we ask three questions:

1. **What is the composition of each Geomarket?** To what extent do College Board Geomarkets partition academically similar prospects into racially and socioeconomically distinct pools within metropolitan areas?

2. **Where does each group come from?** Which Geomarkets contribute disproportionately to different racial/ethnic and first-generation-status groups within academically similar prospect pools?

3. **What happens when a Geomarket is excluded?** How does excluding particular Geomarkets change the composition of the prospect pool made visible to colleges?

We answer these questions using actual student lists purchased by public universities, which we obtained through public records requests. We digitized College Board Geomarket maps published in College Board (2023a) into GIS shapefiles and assigned prospects to Geomarkets. By analyzing purchases that include all Geomarkets within a metropolitan area, we reconstruct counterfactual lists and vary Geomarket inclusion while holding academic criteria constant. We present detailed analyses of the Chicago metropolitan area and compare the results with those for other metropolitan areas in the Online Appendix. Chicago-area Geomarkets

differ sharply in racial and first-generation composition, yet each contains large numbers of high-achieving students. Geomarkets therefore partition academically similar prospects into racially and socioeconomically distinct pools, making the representation of particular groups depend on which territories are included in a purchase. For example, excluding the "IL 11 – City of Chicago" Geomarket would disproportionately reduce the representation of Black, Hispanic, and first-generation Asian prospects. Across metropolitan areas, filtering toward affluent Geomarkets and away from poorer ones can disproportionately exclude Black, Hispanic, and first-generation Asian students from colleges' prospect pools.

These analyses contribute to scholarship on college access by moving inquiry upstream from observable college behavior to the third-party products that make particular forms of organizational behavior possible. The Geomarkets case shows how an intermediary translates social structure into organizationally usable categories. College Board did not create residential stratification, and Geomarkets do not dictate colleges' choices; rather, College Board standardized one representation of unequal space and embedded it in a product that makes geographically differentiated prospect selection possible. Products neither dictate institutional behavior nor neutrally execute preexisting preferences; their consequences emerge as product affordances interact with organizational priorities and established practices. More broadly, this case motivates a sociology of vendor-mediated education that treats commercial infrastructures as consequential sites through which inequality becomes actionable.

2 Literature and Case

2.1 Literature Review

College access. Sociology explains inequality in college access through two broad traditions. The status-attainment tradition argues that educational attainment and college destinations are strongly patterned by family socioeconomic background, especially parental education, occupation, and income (Blau and Duncan 1967; Sewell and Shah 1968). A second tradition,

drawing on Bourdieu's (1984) theory of cultural capital and Weberian accounts of status-group competition (Collins 1994), emphasizes how privileged households and elite institutions reproduce advantage (Huang 2023; Lee and Kramer 2013; Karabel 2005). A related body of scholarship examines how college behavior and processes shape access for students. This literature focuses on selective admissions processes (e.g., holistic admissions, test-optional) (Hsu and Sharkey 2025; Bastedo et al. 2016) and to a lesser extent financial aid (McPherson and Schapiro 1998; Doyle 2010), and is often set against macro institutional forces such as college rankings, affirmative action, and college entrance exams (e.g., Alon 2009).

A growing literature on recruiting examines how colleges identify and solicit the prospective students they hope to convert into applicants. Ethnographies by Stevens (2007) and Khan (2011) identify connections between private school guidance counselors and college admissions officers as a mechanism for social reproduction. In turn, recruiting visits to high schools are a means of maintaining ties with guidance counselors at feeder schools (Stevens 2007; Ruffalo Noel-Levitz 2022b). Quantitative case-studies of off-campus recruiting visits by public research universities and by selective private universities reveal a preference for visiting private schools and affluent, predominantly White public schools (Author XXXXb, XXXXe, XXXXa). From the student perspective, Holland (2019) finds that underrepresented students were drawn to colleges that made them feel wanted, often attending institutions with lower graduation rates and requiring larger loans than other college options. Cottom (2017) shows that for-profit colleges found a niche in Black and Latinx communities because traditional colleges ignored these communities. Collectively, this literature examines where colleges direct recruiting efforts and how students respond, largely taking for granted how the population available for recruitment is constructed.

However, analyzing recruiting as college behavior obscures the commercial infrastructures that structure which students are targeted by which colleges. College recruiting is shaped by the broader enrollment management industry, including third-party servicers (e.g., College

Board and EAB) that supply databases, classifications, software, and consulting services. Collectively, these products do not merely execute colleges' preferences; they help define the options available to colleges and the populations on which they can act. Scholarship on college access has largely ignored the role of third-party servicers (but see Huang 2023). One important exception is Hamilton et al. (2024). Their analysis of contracts between public universities and online program managers shows how outsourcing recruitment can create incentives to target marginalized students for potentially extractive online programs.

Student list products. Lead generation is the process of converting "prospects" – consumers a business wants to target – into "leads" – prospects whose contact information has been obtained (Federal Trade Commission 2016). In enrollment management, "student lists" (also called "names") are spreadsheets that contain the contact information of prospects (Belkin 2019). Student lists are a fundamental input for undergraduate recruiting campaigns because purchased names – alongside prospects who reach out on their own – constitute the set of prospects who receive subsequent recruiting interventions (e.g., mail, email) designed to push them to apply and enroll. Student list products are "selection devices" – routines for making decisions about who receives an intervention (Hirschman and Bosk 2020) – that enable colleges to select which names they purchase by incorporating search filters (e.g., HS graduating class, state, test score). For example, a college might select all California prospects from the 2028 HS class, with a GPA of 3.5+, and an SAT score between 1200-1300. Historically, the largest student list vendors are College Board and ACT, which create student list products based on their database of test-takers. In 2021, the College Board Student Search Service charged $0.50 per name (College Board 2021).

Ruffalo Noel-Levitz (2022b) reports that 86% of public colleges and 87% of private colleges purchase student lists. For public universities that purchased lists, 80% purchase more than 50,000 names annually. Ruffalo Noel-Levitz (2022a) reports that student lists were the top expenditure item in the undergraduate recruiting budget for both private and public

institutions in 2022. Prior research finds a positive relationship between student lists and student outcomes [Howell et al. (2021)]. After controlling for SAT scores, parental education, and school fixed effects, 41.1% of SAT test-takers who opted into Student Search Service – thereby allowing colleges to purchase their contact information – attended a four-year college, compared with 32.8% of those who opted out (Howell et al. 2021).

However, Author (XXXXc) argue that student list products are selection devices that exacerbate racial inequality in college access. Norris (2021) defines "racialized inputs" as ostensibly race-neutral inputs that are systematically correlated with race because marginalized racial/ethnic groups have historically been excluded from the input. Several frequently utilized student list filters (e.g., zip code, AP test score) meet the criteria of racialized inputs. For example, colleges may purchase prospects that achieve a certain score on a certain AP test, but Black and Hispanic students are more likely to attend high schools with fewer AP offerings (Rodriguez and McGuire 2019). Using a national sample of high school students and also data from actual student lists purchased by public universities, Author (XXXXc) show that racialized search filters have a strong negative relationship with the selection of Black and Latinx prospects.

Student Search Service offers filters based on familiar geographic boundaries as well as proprietary geographic classifications created by College Board. These include Geomarkets, which parcel metropolitan areas into smaller recruiting territories, and Geodemographic Segment, which groups census tracts and high schools into clusters based on college-going behavior (College Board 2011). When commercial intermediaries transform metropolitan space into market categories, they package historically produced inequalities into actionable units for organizational comparison and selection. Korver-Glenn (2022) shows how racially segregated "market areas," developed by the Houston Association of Realtors, contribute to lower home value appraisals in predominantly Black and Hispanic neighborhoods. This manuscript investigates which prospects are included or excluded when student list purchases

filter on particular Geomarkets, thereby showing how vendor-created categories can change the racial and class composition of colleges' prospect pools. The contribution to the sociology of college access is to move the object of empirical inquiry upstream, from college behavior itself to the products that make college behavior possible.

2.2 The Case: Geomarkets and The Market Segment Model

In 1978, set against the backdrop of impending college-age demographic decline, University of Pennsylvania professor Robert Zemsky was tasked by the president to figure out, "Who thinks about Penn?" and "What other institutions do they think about when they think about us?" (Zemsky and Oedel 1983, x). To answer these questions, Zemsky began working with the Market Research Committee of the Consortium on Financing Higher Education (COFHE), a consortium of 30 selective private institutions. Zemsky and Oedel (1983, x) write, "Members of COFHE had come to know almost as much about one another's applicant pools as they did about their own." The project goal became the creation of the Market Segment Model, which had the ambition of predicting student demand for any college from any locality. However, "to gain a truly comprehensive view of the collegiate enrollment market, we needed a database that described most institutions and most students" (Zemsky and Oedel 1983, x). The group approached College Board. "Coincidentally, the Board was reviewing its own efforts to help colleges estimate their enrollment potential" (Zemsky and Oedel 1983, x). College Board underwrote the project, providing data and funding (Zemsky and Oedel 1983).

Zemsky and Oedel (1983) describe the Market Segment Model as an effort to "capture and quantify" the knowledge of admissions officers [p. 11]. They write that "the intuitions of admissions officers actually comprise a remarkably systematic body of knowledge about the college selection process" and that "our research is based upon listening carefully to what admissions officers have to say" (Zemsky and Oedel 1983, 9–10). The "initial task" was to define geographic boundaries "in a manner consistent with admissions officers' intuitive understanding of student pools" (Zemsky and Oedel 1983, 4). Quoting an admissions officer,

Zemsky and Oedel (1983) [p. 11] write, "There are only three kinds of college-bound students: those who want to live at home, those who want to live on campus but bring their laundry home, and those who want to go far enough from home that Mom and Dad can't visit without calling first." As such, Zemsky and Oedel (1983, 11) created "three types of boundaries—region, state, and community." These community-based enrollment markets were later called Geomarkets. In metropolitan areas, "each major metropolitan area is composed of several markets, usually corresponding to the inner city, a first ring of suburbs, and an outer ring of suburbs" (Zemsky and Oedel 1983, 11–12).

The Market Segment Model is a classification system that groups SAT test-takers into one of four different *market segments*–local, in-state, regional, and national–based on SAT score-sending behavior. For a given student, each college receiving a score could be defined as "local" (located in the same Geomarket as the student), "in-state" (same state but different Geomarket), "regional" (same region but different state), or "national" (different region). In turn, a test-taker was categorized in the local market segment if they submitted more SAT scores to local institutions than to institutions in any other geographic category. Zemsky and Oedel (1983, 4) describe the model as "nothing more than a set of simple rules for disaggregating high school seniors into similar groups," explaining that "the model worked because students, once so disaggregated, appeared to behave in remarkably consistent ways."

Zemsky and Oedel (1983) identify four variables—parental education, family income, scholastic aptitude, and educational aspirations—that predicted whether a student belongs to the local, in-state, regional, or national market segment. The resulting thesis is that student demand for college is a function of social origin. The authors state that "local and in-state students are not likely to come from families in which both parents have received college educations" (Zemsky and Oedel 1983, 33). Commenting on family income, Zemsky and Oedel (1983, 33) write that "we could predict that all local students would come from moderate-income or low-income families and be wrong only 5.5 percent of the time." They conclude that "our research has

simply demonstrated what everyone has always known: communities with high levels of family income and parental education are also communities in which students have higher than average SATs and more far-reaching aspirations" (Zemsky and Oedel 1983, 42). However, Geomarkets differ in the relative abundance of students from particular socioeconomic origins, which has practical implications for recruiting.

Zemsky and Oedel (1983) recommend that colleges target Geomarkets with desirable compositions of socioeconomic characteristics in order to reach students from desired market segments. A community college should concentrate recruiting efforts on students in the local market segment who live in the same Geomarket as the college. Regional state colleges and non-selective private colleges should focus on students from the local and in-state segments. Selective private colleges and public universities looking for out-of-state students should focus on students from the regional and national segments. For a college drawing most of its students from these latter segments, the authors advise, "Ideally, you would seek communities with a high proportion of students already predisposed toward institutions such as your own. The Market Segment Model would provide this information through segment percentages for the community in question" (Zemsky and Oedel 1983, 44). They acknowledge that "On occasion, senior spokespersons for the profession worry that students outside the main [Geo]market areas remain forgotten and hence, unchallenged. Inevitably, the increasing competition for students, the expense of travel and mailings, and internal political constraints compel institutions to concentrate their efforts where they will do the most good. The result is a natural reinforcing of the basic socioeconomic patterns that gave shape in the first place to the structure of college choice" (Zemsky and Oedel 1983, 44).

College Board developed commercial enrollment products based on projects by member colleges that College Board underwrote. For example, support for Carleton College's landmark market research study yielded the Admitted Student Questionnaire product (Litten et al. 1983). In 1984, College Board created the Enrollment Planning Service (EPS), based on

Geomarkets and the Market Segment Model (College Board 2012; Takamiya 2005). EPS recreated the analyses of Zemsky and Oedel (1983) through reports produced separately for each Geomarket. The Market Segment Profile provided information about the number and characteristics of students in each market segment. The Institutional Profile showed the number of students who sent scores to a particular institution, separately by market segment; colleges could obtain this information for their own institution or for a competitor. EPS also provided information about the score-sending behavior of individual high schools within each Geomarket (College Board 2019). Colleges could use these reports to identify which Geomarkets to recruit from and which high schools to visit within those territories. College Board marketed EPS as "the marketing software that pinpoints the schools and Geomarkets where your best prospects are most likely to be found. With the click of a mouse, EPS provides you with comprehensive reports on your markets, your position in those markets, and your competition. Focus your valuable time and resources on the right prospects" (College Board 2005).

Whereas Zemsky and Oedel (1983) identified 143 Geomarkets covering New England, Middle States, and the South, EPS created Geomarkets for the remaining U.S. states, with 304 Geomarkets in total (College Board n.d.). Software documentation by Oracle (n.d.) states that "EPS market codes are proprietary market codes owned by the College Board and are used to categorize external organizations and people into geographical areas, mostly in the United States." Noel-Levitz (1998) reports that in 1995, 37% of four-year public institutions and 49% of four-year private institutions used EPS.

Geomarkets remain part of contemporary admissions practice and enrollment-management software. Admissions professionals describe recruiting assignments using numbered Geomarkets or EPS markets, while college admissions websites show that admissions recruiters are assigned to territories that correspond to College Board Geomarkets (Warren 2025; Porter 2026; Montecalvo 2026; Cleland 2026; College Board 2023b; University of Chicago, n.d.; Lake

Forest College 2026; Saint Mary's College 2024). Slate – a customer relationship management (CRM) platform used by more than 2,000 colleges and universities (Technolutions, n.d.) – supports Geomarket-based staff assignments, territory reporting, and travel planning, and recommends Geomarket filters over lists of individual school codes (Technolutions 2026a, 2026c, 2026b, 2026d). Demonstration screenshots also show Geomarket labels on individual and high-school records and Geomarket filters within Slate's query and rule interfaces (Anonymous 2026). Similarly, the Oracle PeopleSoft CRM supports loading College Board's EPS market codes and assigning them to external organizations (Oracle n.d.).

Student Search Service extended the Geomarket/EPS architecture from market analysis to prospect selection. College Board documentation confirms that by 2001 colleges could purchase prospect profiles by filtering on particular Geomarkets (College Board 2001), although we suspect this integration occurred earlier. Anticipating this integration, Zemsky and Oedel (1983, 42–44) describe a hypothetical selective college that pursues its enrollment goals by filtering on affluent Geomarkets when deciding which prospect profiles to purchase. They explain that "Further classification of students by social attributes allows you to identify a group for mailings" (Zemsky and Oedel 1983, 44). College Board marketed EPS as a complement to Student Search Service, describing it as "an analysis and reporting service that enables admission professionals to better understand the student market and the lists they license with Student Search Service" (College Board 2022).

3 Theoretical Framework: Market Devices

Economic sociology challenges atomized accounts of economic action by emphasizing that markets are socially constructed. The "structural embeddedness" paradigm argues that markets are a consequence of interpersonal connections (Granovetter 1985; Fourcade 2007). By contrast, the "market devices" literature shifts attention toward the technologies that make market behavior possible. Rather than treating technologies as neutral tools operating

within socially structured markets, market devices scholars examine how calculative devices, classifications, and other material arrangements participate in constituting markets and the actors who operate within them (Callon 1998; Fourcade 2007). That is, markets do not exist a priori waiting to be observed; they are constructed by market devices (Muniesa et al. 2007).

Market devices are tools for calculating, comparing, and classifying (e.g., algorithms, metrics, rankings) that become incorporated into products and transactions "that make economic action possible" (Hirschman and Garbes 2019, 1176). Muniesa et al. (2007, 2) define market devices as a "way of referring to the material and discursive assemblages that intervene in the construction of markets." An assemblage refers to a bundle of heterogeneous components that work together, suggesting that market devices consist of and build upon other market devices. Drawing from Krippner's (2024) study of insurance, examples of market devices are the statistical models that assign individuals to "risk pools" for insurance products. Market devices shift scholarly attention to the individual components—including techniques, theories, cultural logics, and histories—that are foundational to products and to markets (Fourcade 2011).

Market devices shape what actors can do and what they can see. The "affordances" of a market device are the practical actions the device makes possible, including built-in options, settings, and outputs that enable users to do some things more easily than others. "Framing" refers to how the device constructs the decision situation, including what is rendered visible, eligible, and meaningful for action. For example, Pollock and D'Adderio (2012) examine Gartner's Magic Quadrant, a diagram used by IT managers to decide which prospective software vendors to consider. Its standardized dimensions afford comparison, while its inclusion rules and categories frame which vendors appear and how they are evaluated. Similarly, enrollment management (EM) managers use Student Search Service filters to decide which prospective students to target. These filters afford selection through predefined classifications and frame who appears on purchased lists. In both cases, the market device structures decision-making

by enabling particular actions and defining the alternatives visible for consideration.

In the market devices framework, neither market devices nor actors have agency in isolation. "Agencement" refers to an arrangement of market devices, actors, practices, and organizational goals and pressures that creates the capacity for action in a particular context (Çalışkan and Callon 2010). This arrangement makes some courses of action available, recognizable as choices, and more likely than others. For example, in the contemporary "market configuration" of higher education (Stevens and Gebre-Medhin 2016), defined by state disinvestment and competition for paying customers, university enrollment goals may prioritize "full-pay" students. Within this agencement, EM managers can use Student Search Service filters to purchase prospect lists concentrated in affluent communities and direct recruitment communications toward those students (Zemsky and Oedel 1983).

Integrating the embeddedness and market devices literatures, market devices emerge from the collective action of socially embedded networks of people and organizations. MacKenzie and Millo (2003) show that the Chicago Board Options Exchange emerged through collective action within a close-knit trading community. Repeated interaction, shared cultural commitments, and norms of reciprocity and mutual sanctioning sustained members' contributions to a project vulnerable to free-riding. Similarly, Müller-Seitz and Sydow (2012) trace how Intel and other powerful firms shape the International Technology Roadmap for Semiconductors through the SEMATECH industry network. The roadmap functions as a market device by translating priorities negotiated among these firms into technological targets that guide development and investment decisions across the industry. Once incorporated into commercial products and practices, these arrangements become available to users beyond the groups that created them, carrying their categories and operating assumptions into new settings.

3.1 Application to the Case

Our application traces the sequence connecting the social origins of market devices to the distributional consequences of their use. Geomarkets and the Market Segment Model were conceived from elite colleges' shared understanding of stratified student markets. College Board commercialized and circulated these classifications through its recruiting products, making them available to colleges beyond the originating network. Incorporated into Student Search Service as filters, Geomarkets became options for including or excluding prospective students from recruitment consideration. Our empirical analyses focus on these selection affordances and the racial and socioeconomic consequences of particular Geomarket selections.

Geomarkets were conceived within a network of elite colleges. Discussing their work with COFHE's Market Research Committee, Zemsky and Oedel write: "Much of this project's framework – the Market Segment Model and its derivation from admissions officers' folklore – was shaped and then reshaped in our discussions with members of the committee" (Zemsky and Oedel 1983, x). This elite-college understanding of recruiting territories was applied to metropolitan spaces already structured by substantial racial and socioeconomic inequality (Reardon and Bischoff 2011; Korver-Glenn 2022). The 1980 panels of Figure 2 and Figure 3 document substantial income and racial differences among Chicago-area Geomarkets around the time the classification was developed [CRYSTAL – ADD CLICKABLE LINKS TO THE INTERACTIVE MAPS FOR A FEW MORE GEOMARKETS; LET'S SAY DMV; LOS ANGELES; HOUSTON]. Moving from description to recruiting prescription, Zemsky and Oedel (1983, 44) recommend that colleges target Geomarkets with desirable socioeconomic compositions to reach students from their desired market segments.

College Board is an intermediary that draws on elite colleges' understandings of student college choice to create commercial market devices – EPS and Student Search Service – that mediate between stratified geography and organizational action. Once formalized and embedded in these commercial market devices, the categories no longer depended on the

original admissions community that produced them; they became standardized affordances through which colleges could act on stratified geography to pursue enrollment goals.

For a college representative interested in recruiting prospects from a particular metropolitan area, the Geomarket filter affords the ability to select any Geomarket or combination of Geomarkets. The action the Student Search Service device allows is a student list purchase. The outcome is a spreadsheet that contains one row for each prospective student who met all search criteria. The design of the Student Search Service product frames the decision by showing the college representative only the outcome associated with the chosen set of search filters, not the counterfactual outcomes associated with filters they could have selected (Author XXXXd). Consequently, the purchased list does not show socioeconomic and racial differences between the selected prospects and the prospects who would have been included under alternative Geomarket selections. The market device thus obscures the distributional consequences of actions by leaving counterfactual outcomes outside the decision-making frame.

Our overarching research question is, how do Geomarket filters within College Board's Student Search Service shape the racial/ethnic and first-generation composition of academically similar prospect pools made visible to colleges? Our analyses create a "counterfactual affordance map" that describes the racial/ethnic and first-generation composition of prospect profiles that would have been purchased under particular Geomarket selections. Holding academic criteria constant, our analyses vary the Geomarket filter to show how different filter choices would change the composition of the visible prospect pool.

RQ1 asks, do Geomarkets partition academically similar prospects into racially and socioeconomically distinct pools? Geomarkets classify students geographically, but not explicitly by race/ethnicity or first-generation status. The analysis also shows the prospect pool generated by filtering on a single Geomarket.

RQ2 asks, which Geomarkets contribute disproportionately to different racial/ethnic and

first-generation-status groups within academically similar prospect pools? If, for example, first-generation Asian prospects in a metropolitan area disproportionately come from one or two Geomarkets, then their visibility to colleges depends on whether those Geomarkets are included in the student list purchase.

Finally, RQ3 asks, how does excluding particular Geomarkets change the composition of the prospect pool made visible to colleges? This question examines the device's filtering affordance by estimating how the visible prospect pool changes when a particular Geomarket is excluded from a student list purchase.

College recruiting operates through an agencement consisting of organizational goals and pressures, professional associations (e.g., NACAC), recruiting routines, and market devices such as Geomarkets, EPS, and Student Search Service. The enrollment management industry is oriented toward tuition revenue and, for a subset of colleges, academic prestige. These priorities orient how enrollment managers use the devices to compare recruiting territories and select prospect populations. This logic is consistent with evidence that selective private colleges and the out-of-state recruiting efforts of public research universities focus disproportionately on affluent, predominantly White schools (Stevens 2007; Author XXXXa, XXXXb; Khan 2010; Quillian and Phillips 2023). Within this agencement, Student Search Service makes class- and race-differentiated territories actionable by allowing colleges to translate enrollment goals into geographically defined prospect pools.

4 Data and Methods

4.1 Research Design and Case Selection

Since Geomarkets partition metropolitan space into proprietary recruiting territories, we utilize a quantitative case study design in which metropolitan areas are the cases. Our empirical goal is to reconstruct how Geomarket filters within the College Board Student

Search Service shape the racial/ethnic and first-generation composition of academically similar prospect pools made visible to colleges. We do so by analyzing actual student-list purchases and varying Geomarket inclusion and exclusion while holding academic search criteria constant. Consistent with our conceptual framing, Student Search Service is treated as a market device whose built-in filters define a menu of possible actions. The analyses therefore focus on the distributional consequences of one filtering affordance built into the product.

Our research questions require observing all Geomarkets within a metropolitan prospect pool. We therefore restrict analyses to metropolitan areas for which we have student-list purchases that (A) include every Geomarket, (B) define prospect pools primarily by graduating class and test-score range, and (C) repeat that structure across low, middle, and high test score bands. The primary analyses focus on prospects in the middle test score band, with results for low and high bands in the Online Appendix. We exclude purchases that layer on additional search criteria, like intended major or preferred college size, that produce dramatic and non-random reductions in prospect pools. The final sample includes the Bay Area, Chicago, Dallas, Detroit, Long Island, Los Angeles, Northern New Jersey, Orange County, Philadelphia, and San Diego.

Given space constraints, we present Chicago as the primary case because its seven Geomarkets follow the canonical metropolitan structure described by Zemsky and Oedel (1983) while exhibiting substantial racial and socioeconomic differentiation. This allows readers to observe the relationship between Geomarket composition (RQ1), group distributions (RQ2), and exclusion consequences (RQ3) within a single metropolitan structure. Results for all metropolitan areas are reported in the Online Appendix, but we provide a brief comparative discussion of high-level results across all metropolitan areas in the sample. We emphasize that Chicago is presented as an illustrative case rather than a representative metropolitan area.

4.2 Data

The primary data consist of student lists purchased through the College Board Student Search Service by public universities. We obtained these data through public records requests submitted to public universities in California, Illinois, Texas, and Arizona. The broader project dataset this manuscript draws from includes a total of 414 orders associated with 2,549,085 prospects made by 11 public universities from 2016 through 2020. Although lists were purchased by individual universities, the resulting prospect pools are a function of the search criteria specified for that student list purchase. A different university that selected the same search filters would obtain the same resulting prospect profiles, except where universities opted to exclude prospect profiles previously purchased.

For each student list purchased, we requested two related pieces of data: (1) the order summary specifying search criteria; and (2) the de-identified prospect-level list produced from those criteria. Prospect-level records contain information about students' high schools, geographic location, race/ethnicity, parental education, and other variables included in the purchased list. We use parental education to construct a first-generation indicator.

Although this study does not aim to estimate the prevalence of Geomarket filter usage, our records request data provide some insight into how it is used in practice. Of the 414 orders in the project dataset, approximately 13% (N=52) used a Geomarket filter. These orders spanned multiple universities, years, academic thresholds, and metropolitan areas throughout the U.S. Records also suggest Geomarket filters were sometimes used to distinguish affluent communities from lower income ones within the same metropolitan area, but the frequency of this practice varied across locations. For instance, 39 of the 52 order purchases using Geomarket filters targeted Chicago. Almost 80% (N=31) of these orders included one or more affluent suburban Geomarkets while excluding the lowest-income City of Chicago Geomarket. Similarly, orders targeting the DC/Maryland/Virginia metropolitan area predominantly selected for more affluent Geomarkets (e.g., Montgomery Metropolitan;

Arlington & Alexandria; Fairfax County; Central Maryland (w/o Baltimore)) while excluding poor, substantially Black Geomarkets (e.g., Prince George's Metropolitan; Baltimore (Urban)). Although we do not claim that such data represent how all colleges use Geomarkets, such patterns indicate that selective filtering across Geomarkets is neither isolated nor incidental. These patterns support our conceptualization of Geomarket filters as selection devices that provide various affordances for differential inclusion and exclusion, rather than serving as descriptive reporting regions within EPS.

We also incorporated data from the 1980 Decennial Census, the 2000 Decennial Census, and the 2020 ACS 5-year estimates to provide descriptive historical and contemporary context about the racial and socioeconomic composition of Geomarkets.

Geomarket boundaries and prospect assignment

We digitized College Board Geomarket maps published in College Board (2023a) into GIS shapefiles. To assign each Census tract to a Geomarket, we georeferenced the College Board's Geomarket maps by overlaying PDF images onto spatial coordinates aligned with census tract boundaries. We then extracted vector data by manually tracing the Geomarket boundaries and generating individual shapefiles for each Geomarket. This approach aligned precisely with 2020 census tract boundaries from ACS 5-year estimates; that is, each census tract corresponded uniquely to a Geomarket generated from the College Board images. Finally, we spatially assigned prospect records to Geomarkets based on the geographic coordinates of their high school.

4.3 Analyses

We utilize descriptive statistics, tables, and maps to analyze how Geomarket filters structure the composition of visible prospect pools. First, we provide contextual descriptive analyses of Geomarkets themselves by describing the racial and socioeconomic differentiation across Geomarkets from 1980, 2000 and 2020.

Analyses for RQ1-RQ3 rely on student list purchase data. For RQ1, we examine the racial/ethnic and first-generation composition of academically similar prospects – based on test score thresholds – within each Geomarket. These analyses show how Geomarket filtering partitions academically similar students into prospect pools that differ by race/ethnicity and parental education. For RQ2, we calculate the share of each racial/ethnic and first-generation group contributed by each Geomarket within a metropolitan prospect pool. These analyses identify Geomarkets that disproportionately contribute particular groups to the overall pool of academically eligible prospects. For RQ3, we show how the racial/ethnic and first-generation composition of the prospect pool changes when specific Geomarkets are excluded from a student list purchase. These analyses estimate which groups become disproportionately invisible under alternative Geomarket selections.

5 Limitations

Our primary limitation is geographic coverage. Analyses require student list purchases that encompass all Geomarkets within a metropolitan area. Without such purchases, we cannot reconstruct the full metropolitan prospect counterfactual pool or estimate how the pool changes when individual Geomarkets are excluded. We further restrict analyses to metropolitan areas where we have purchases across a range of test-score thresholds and where purchases do not condition on filters beyond high school graduation year, GPA, test-score threshold, and geography. Consequently, this reduces the study's case sample to the ten metropolitan areas listed above. Results should be interpreted as case-based evidence of how the Geomarket filter operates in metropolitan areas we analyze, not as nationally representative estimates of Geomarket filtering across all U.S. metropolitan areas.

6 Results

6.1 Geomarket socioeconomic and racial context

Figure 1 provides a static map of the seven "Greater Chicago" Geomarkets, as defined by College Board (2023a). The Online Appendix provides an interactive version, showing change over time in population, race/ethnicity, poverty, and educational attainment at the tract- or Geomarket-level. The Geomarkets vary substantially in population size. In 2020, populations range from about 161,000 in North Shore to 3.2 million in the City of Chicago, with Western Suburbs (1.4 million) and South & Southwest Suburbs (1.2 million) also substantially larger than Evanston & Skokie (309,000).

Figure 2 shows Geomarkets also vary substantially by socioeconomic characteristics. Focusing on 2020, North Shore has the highest median income, followed by Evanston & Skokie, Western Suburbs, Northwest Suburbs, and Chain of Lakes. At the opposite end, the City of Chicago has the lowest median income and highest poverty rate, followed by the South & Southwest Suburbs, which has the lowest share of population with a bachelor's degree or higher.

Figure 3 shows similar sharp variation across racial/ethnic composition. By 2020, City of Chicago remained the most diverse Geomarket, with no racial/ethnic group constituting a majority, while North Shore remained comparatively White. Since 1980, several predominantly White suburban Geomarkets became more racially heterogeneous, especially the Chain of Lakes, Northwest Suburbs, Western Suburbs, and South and Southwest Suburbs. This change reflects growth in Hispanic populations, alongside increases in Black residents in the South and Southwest Suburbs and Asian residents in the Northwest Suburbs and Evanston and Skokie. By 2020, Black and Hispanic residents together accounted for 55.4% of the City of Chicago Geomarket and 43.0% of the South and Southwest Suburbs.

6.2 Research Question 1

RQ1 asks the extent to which College Board Geomarkets partition academically similar prospects into racially and socioeconomically distinct pools within metropolitan areas. Figure 4 shows the racial/ethnic composition of purchased student profiles by Geomarket across three different SAT score ranges: 1020 - 1150, 1160 - 1300, and 1310 - 1600. Focusing on the middle test score range, the racial/ethnic composition of prospect pools varied sharply across Chicago-area Geomarkets. White prospects were concentrated in North Shore, Evanston & Skokie, and Western Suburbs, while Asian prospects were concentrated in Evanston & Skokie, Northwest Suburbs, and North Shore. Black and Hispanic prospects were concentrated most heavily in the City of Chicago and South & Southwest Suburbs. Asian representation increases especially sharply in North Shore and Western Suburbs at the highest score range, however these broad racial differences in Geomarket composition persist across all score ranges.

Figure 5 shows similarly large differences in first-generation composition across Geomarkets, with the share of first-generation students generally declining as test score thresholds rise. In the middle score range, first-generation prospects comprise approximately 47% of profiles in the City of Chicago and 38% in South & Southwest Suburbs, compared to 22% in Western Suburbs, 19% in Evanston & Skokie, and 11% in North Shore. Northwest Suburbs (29%) and Chain of Lakes (24%) fall between these ranges. Such patterns reflect broader population socioeconomic differences described above, as Geomarkets with lower median income, higher poverty, and lower educational attainment contain substantially larger shares of first-generation prospects.

Figure 6 presents first-generation status by race for the middle test score range for White, Asian, Black, and Hispanic students. The figure shows first-generation status varies substantially across Geomarkets even within the same racial/ethnic group. These within-race differences are largest for Asian and Hispanic prospects and smaller for White and Black

prospects. Among Asian prospects, more than 50% of students from the City of Chicago are first-generation, compared to roughly 20% in Western Suburbs. Among Hispanic prospects, approximately 75% of students from the City of Chicago are first-generation, compared to about 25% in North Shore. Together, results indicate Geomarkets separate not only racial/ethnic groups from one another; they also distinguish socioeconomically different subsets within racial/ethnic groups.

6.3 Research Question 2

RQ2 asks which Geomarkets contribute disproportionately to different racial/ethnic and first-generation groups within academically similar prospect pools. Figure 7 shifts attention from the composition of individual Geomarkets to their contributions to the overall Chicago metropolitan prospect pool. The lollipop lines show each Geomarket's share of a given racial/ethnic group, while the grey bars show its share of all prospects; lines longer than their corresponding grey bars indicate overrepresentation, and shorter lines indicate underrepresentation.

White prospects are disproportionately drawn from suburban Geomarkets, especially Western Suburbs, South & Southwest Suburbs, Chain of Lakes, and Northwest Suburbs, while City of Chicago contributes disproportionately fewer White prospects. Asian prospects are concentrated in Western Suburbs, City of Chicago, Northwest Suburbs, and Evanston & Skokie, with North Shore also contributing disproportionately more Asian prospects. Black prospects are disproportionately concentrated in City of Chicago and South & Southwest Suburbs, with affluent suburban Geomarkets contributing disproportionately fewer Black prospects. Hispanic prospects are similarly concentrated in the City of Chicago while also drawing substantial contributions from Western Suburbs, South & Southwest Suburbs, Chain of Lakes, and Northwest Suburbs. Overall, Figure 7 shows selecting or excluding a Geomarket changes not only the size of the prospect pool but also which racial/ethnic groups are disproportionately made visible or invisible within that pool.

Figure 8 shows similar patterns by first-generation status. The City of Chicago stands out as an engine for first-generation prospects, contributing 23% of all prospects but 35% of first-generation prospects. Similarly, South & Southwest Suburbs contributes 17% of all prospects but 21% of first-generation prospects. By contrast, North Shore and Western Suburbs contribute disproportionately more non-first-generation prospects relative to their overall share of the metropolitan prospect pool.

Figure 9 compares each Geomarket's contribution to the overall prospect pool (gray bars) with its contribution to first-generation (blue lines) and non-first generation (red lines) within each racial/ethnic group. Differences between the gray bars and colored lines indicate where race-by-first-generation subgroups are over- or underrepresented relative to the overall prospect pool. Among White prospects, first-generation students are drawn more heavily from the South & Southwest Suburbs, while non-first-generation students are drawn more heavily from the Western Suburbs. City of Chicago, by contrast, contributes a smaller share of both first-generation and non-first-generation White prospects than it does of the overall prospect pool. Among Black prospects, both first-generation and non-first-generation students are concentrated overwhelmingly in the City of Chicago and South & Southwest Suburbs, with relatively small contributions from the remaining Geomarkets.

Asian prospects exhibit a different pattern. First-generation Asian prospects are drawn especially heavily from the City of Chicago, which contributes a much larger share of first-generation Asian prospects than of the overall prospect pool. Non-first-generation Asian prospects, by contrast, are drawn more heavily from the Western Suburbs and Northwest Suburbs. This pattern is not uniform, however; Evanston & Skokie contributes disproportionately larger shares of both first-generation and non-first-generation Asian prospects than it does of the overall prospect pool. Hispanic prospects show a similarly sharp first-generation split. First-generation Hispanic prospects are concentrated especially heavily in the City of Chicago, while non-first-generation Hispanic prospects are more broadly distributed across

suburban Geomarkets.

Overall, Figure 9 suggests that the relationship between Geomarket and first-generation status differs across racial/ethnic groups. For Black prospects, the City of Chicago and South & Southwest Suburbs contribute disproportionately large shares of the overall Black prospect pool, but first-generation and non-first-generation students are distributed similarly across Geomarkets. In other words, Black students' representation is strongly tied to particular Geomarkets, but those Geomarkets do not substantially distinguish first-generation from non-first-generation prospects. Among Asian and Hispanic prospects, first-generation status is more sharply geographically patterned, with first-generation students drawn heavily from the City of Chicago and non-first-generation students drawn from suburban Geomarkets. White prospects fall between these patterns: first-generation and non-first-generation White students are less geographically separated, but South & Southwest Suburbs contributes a larger share of first-generation than non-first-generation White prospects.

6.4 Research Question 3

RQ3 asks how excluding particular Geomarkets changes the composition of the prospect pool made visible to colleges. Figure 10 and Figure 11 estimate these counterfactual exclusion scenarios by race/ethnicity, showing changes in percentage point and relative percent change, respectively. Figure 10 shows White students represent 60.4% of the prospect pool when all Greater Chicago Geomarkets are included. Excluding the Western Suburbs Geomarket reduces White representation by 3.1 percentage points, which is equivalent to a 5% relative decline as shown in Figure 11. By contrast, excluding the City of Chicago increases it by 7.6 percentage points.

By contrast, Black students represent 7.0% of the full prospect pool, but excluding the City of Chicago Geomarket reduces their representation by 2.6 percentage points, equivalent to a 36% relative decline (Figure 11) Similarly, excluding the South & Southwest Suburbs reduces

Black student representation by 0.8 percentage points, from 7.0% to 6.2% of the overall prospect pool, an 11% decline. By contrast, excluding the Western Suburbs would increase Black students' share of the remaining prospect pool by 1.4 percentage points, from 7.0% to 8.4%, a 20% relative increase.

Hispanic students show comparable patterns. Excluding the City of Chicago reduces Hispanic students' share of the remaining prospect pool by 5.2 percentage points, from 18.8% to 13.6%, a 28% decline relative to their full-pool baseline. By contrast, excluding the Western Suburbs would increase Hispanic students' share of the remaining prospect pool by 2.0 percentage points, from 18.8% to 20.8%, a 10% increase relative to their full-pool baseline. Together, results show that excluding the City of Chicago would not merely remove a large Geomarket from the purchase. It disproportionately reduces the visibility of Black and Hispanic prospects while increasing visibility of White prospects.

Figure 12 and Figure 13 repeat the counterfactual analysis by first-generation status. First-generation students comprise 30.5% of the prospect pool across all Geomarkets, but a disproportionate share of these students come from the City of Chicago and South & Southwest Suburbs Geomarkets. Excluding the City of Chicago would reduce the first-generation share by 4.8 percentage points, from 30.5% to 25.7% of the remaining pool, or a 16% relative decline (Figure 13) Similarly, excluding South & Southwest Suburbs would reduce the first-generation share by 1.5 percentage points, from 30.5% to 29.0%, representing 5% decline relative to the full-pool baseline. Results for non-first-generation students move in the opposite direction. Excluding Western Suburbs or North Shore would reduce the non-first-generation share of the remaining pool, while excluding the City of Chicago or South & Southwest Suburbs would increase it.

Figure 14 and Figure 15 extend the counterfactual analysis to race by first-generation subgroups. Excluding the City of Chicago produces the clearest reduction in visibility for students who are both non-White and first-generation. Relative declines shown in

Figure 15 are greatest for first-generation Black (43%), Hispanic (37%), and Asian (21%) prospects. Excluding the City of Chicago also reduces the relative representation of non-first-generation Black (33%) and Hispanic (13%) prospects, but increases non-first-generation Asian representation by 11%. For White students, however, excluding the City of Chicago yields a 10% relative increase in first-generation representation and a 13% relative increase in non-first-generation representation.

South & Southwest Suburbs offers an interesting contrast. Like City of Chicago, its exclusion generates a relative decline in Black representation for both first-generation (12%) and non-first-generation prospects (10%). Unlike City of Chicago, however, excluding South & Southwest Suburbs also reduces first-generation White representation, producing a 17% relative decline. Western Suburbs provides the clearest counterpoint. Excluding Western Suburbs reduces the relative representation of non-first-generation White prospects and increases the relative representation of several first-generation and racially minoritized subgroups.

6.5 Other metropolitan areas

Metropolitan areas in the Online Appendix, including Philadelphia, Detroit, and Dallas, exhibit Geomarket dynamics similar to Chicago. In each, one or two key Geomarkets reflecting concentrated geographical inequality largely determine the (in)visibility of first-generation prospects across racially minoritized groups. In Philadelphia, excluding central-city Philadelphia County Geomarket, which contains the lowest income and educational attainment communities, reduces the relative representation of first-generation Black (-51%), Hispanic (-42%), and Asian (-37%) prospects[1]. In Detroit, excluding Wayne County similarly reduces first-generation Black and Hispanic prospect representation, whereas excluding the Northern Suburbs Geomarket increases it. Dallas also shows exclusion concentrated in two key Geomarkets. Although the urban-core City of Dallas Geomarket ranks lowest in median income and educational attainment and the more rural Dallas County ranks near the middle,

excluding either reduces the representation of first-generation Hispanic and Black prospects. Together, these metropolitan areas show that where Geomarket boundaries align closely with concentrated racial and socioeconomic divisions, Geomarkets enable colleges to act on clearly delineated inequalities by excluding key Geomarkets central to the visibility of first-generation and racially minoritized prospects.

Unlike Chicago, metropolitan areas like Los Angeles exhibit more dispersed, overlapping, and multicentered patterns of race and class, such that no single Geomarket drives the potential for exclusion across all groups. For instance, excluding the affluent West Los Angeles and West Beach Geomarket produces the same relative reduction of first-generation Black prospects (-9%) as excluding the lower-income South and South Central Los Angeles Geomarket. Yet, excluding South and South Central Los Angeles would also reduce first-generation Hispanic representation, whereas excluding West Los Angeles and West Beach would increase it. Similarly, including East Los Angeles Geomarket is most important for increasing first-generation Asian representation while decreasing first-generation Black representation.

Comparable dynamics appear in the Bay Area for middle PSAT score thresholds. Excluding Contra Costa County reduces first-generation Black and Hispanic representation while increasing first-generation Asian representation. Excluding the City of San Jose increases both first-generation Black and Asian representation but produces almost no change for Hispanic prospects. Similar patterns are evident across the inclusion versus exclusion of Middlesex County, Bergen County, and Essex and Southern Passaic County Geomarkets in the New Jersey metropolitan area. In such polycentric metropolitan areas where racial and class inequalities are not reducible to a single disadvantaged core, Geomarket filters enable affordances for acting on geographical inequalities that do not otherwise appear as clearly bounded or mutually exclusive. By geographically dividing metropolitan areas like Los Angeles, the Bay Area, and New Jersey, Geomarket filters package overlapping inequities

into selectable recruiting options with consequences for who is included versus excluded.

7 Discussion

The analyses show that Geomarkets structure which students become visible to colleges as recruitable prospects. Among academically similar students, Geomarket boundaries separate prospects into racially and socioeconomically distinct pools. Some racial/ethnic and first-generation groups come disproportionately from particular Geomarkets, making their visibility depend on whether those Geomarkets are included in a student-list purchase. In contexts like Chicago, where disadvantage is concentrated within a small number of consequential Geomarkets, excluding poorer Geomarkets sharply reduces the visibility of multiple racially minoritized and first-generation groups. In polycentric areas like Los Angeles, the Bay Area, and New Jersey, disadvantage is spread across several Geomarkets, so excluding one territory has a smaller effect. Yet targeting affluent Geomarkets while excluding poorer ones produces the same overall pattern: racially minoritized and first-generation prospects become less visible.

This manuscript contributes to college access scholarship by moving inquiry upstream from college behavior to the third-party products that make such behavior possible. Geomarkets and the Market Segment Model were conceived from elite colleges' shared understanding of stratified student markets. College Board commercialized and circulated these classifications through its recruiting products, making them available to colleges beyond the originating network. Incorporated into Student Search Service as filters, Geomarkets became options for including or excluding prospective students from recruitment consideration. These products do not simply execute institutional preferences; they help organize the populations and actions on which those preferences operate. Our analysis of proprietary geographic boundaries within Student Search Service shows how a seemingly neutral product feature can shape which students become available for recruitment consideration.

Beyond its contribution to scholarship on college access, our analysis demonstrates the value of a market devices framework for sociology of education by identifying how vendor products, organizational priorities, and established practices can interact to reproduce inequality. Together, these elements constitute an enrollment-management agencement in which products neither dictate organizational behavior nor neutrally transmit institutional preferences (Muniesa et al. 2007; Pinzur and Duterme 2025). Because higher-income, college-educated households are concentrated in particular Geomarkets, the Market Segment Model encourages selective private colleges and public research universities seeking out-of-state students to focus recruiting interventions on those territories (Zemsky and Oedel 1983). This affordance aligns with established recruiting practices that prioritize affluent, predominantly White schools (Stevens 2007; Author XXXXa, XXXXb; Khan 2011). Contrary to Zemsky and Oedel (1983), our analyses show that many students with competitive test scores live outside the "main" Geomarkets. Geography-based filters therefore risk converting unequal residential space into unequal access to college marketing and recruiting interventions.

This case motivates a broader sociology of vendor-mediated education that understands third-party products as integrated stacks of market devices. Whereas research on College Board has focused almost exclusively on student-facing testing products (Rodriguez and McGuire 2019; Goodman et al. 2020), a critical sociology of College Board should interrogate the business-facing enrollment-management products built atop its testing platform. SAT and PSAT scores commensurate students (Espeland and Stevens 1998), and score-sending data make student demand measurable. The Market Segment Model interprets those patterns; Geomarkets classify territory; EPS makes those categories available for market analysis; and Student Search Service turns them into filters for purchasing prospect names. Across the stack, distinct devices collectively shape what can be compared, which markets can be targeted, and which students become visible. Decomposing products into component devices allows researchers to identify which classifications, calculations, filters, interfaces, or integrations create problematic affordances rather than treating entire products as either harmful or benign.

It also helps policymakers identify points at which transparency requirements, evaluation, procurement rules, or other interventions might be applied.

A sociology of vendor-mediated education must situate product stacks within the firms that build them and the financial markets in which they are sold. College Board is an unusual nonprofit in a largely for-profit edtech industry that includes private-equity-backed software firms. Sociology of education has not developed a robust empirical literature on this industry. EAB's Enroll360 illustrates why one is needed. Through acquisitions and partnerships, EAB combined Cappex's proprietary prospect database, Intersect's access to Naviance users, YouVisit virtual tours, and Wisr student communities within a subscription software platform (Author XXXXf; Koppenheffer 2021). Rather than selling names to any college at a per-prospect price, EAB restricts access to these prospect populations to institutions that purchase its software and consulting services. The integrated stack therefore converts controlled access to proprietary prospect populations into a source of recurring revenue and potential institutional dependence. College access is therefore structured not only by the decisions colleges make, but by the third-party products that shape which students become visible and actionable in the first place.

References

Alon, Sigal. 2009. "The Evolution of Class Inequality in Higher Education: Competition, Exclusion, and Adaptation." *American Sociological Review* 74 (5): 731–55.

Anonymous. 2026. *Geomarket Classifications and Filtering in Slate: Demonstration Screenshots*. Unpublished screenshots supplied in confidence.

Author. XXXXf.

Author. XXXXd.

Author. XXXXe.

Author. XXXXa.

Author. XXXXb.

Author. XXXXc.

Bastedo, Michael N., Joseph E. Howard, and Allyson Flaster. 2016. "Holistic Admissions After Affirmative Action: Does 'Maximizing' the High School Curriculum Matter?" *Educational Evaluation and Policy Analysis*, ahead of print. https://doi.org/10.3102/0162373716642407.

Belkin, Douglas. 2019. "For Sale: SAT-Takers' Names. Colleges Buy Student Data and Boost Exclusivity." *The Wall Street Journal*, November. https://www.wsj.com/articles/for-sale-sat-takers-names-colleges-buy-student-data-and-boost-exclusivity-11572976621.

Blau, Peter Michael, and Otis Dudley Duncan. 1967. *The American Occupational Structure*. Wiley.

Bourdieu, Pierre. 1984. *Distinction: A Social Critique of the Judgment of Taste*. Harvard University Press.

Çalışkan, Koray, and Michel Callon. 2010. "Economization, Part 2: A Research Programme for the Study of Markets." *Economy and Society* 39 (1): 1–32. https://doi.org/10.1080/03085140903424519.

Callon, Michel. 1998. "Introduction: The Embeddedness of Economic Markets in Economics." In *The Laws of the Markets*, edited by Michel Callon. Wiley-Blackwell.

Cleland, Mark. 2026. *Admissions Experience at East Carolina University and Becker College*. LinkedIn profile. https://www.linkedin.com/in/mark-cleland-49186b5/details/experience/.

College Board. 2001. *Student Search Help Desk: Selection Criteria*. SearchNOW online user guide. https://web.archive.org/web/20011121021709/http://searchnow.collegeboard.com/16_help11.jsp.

College Board. 2005. *Enrollment Planning Services*. October. Vol. 2005. College Board. www.collegeboard.com/highered/ra/eps.html.

College Board. 2011. *Segment Analysis Service: An Educationally Relevant Geodemographic Tagging Service*. College Board. https://secure-media.collegeboard.org/mSSS/media/pdf/segment-analysis-service-overview.pdf.

College Board. 2012. *Welcome to the College Board*. College Board. https://secure-media.collegeboard.org/homeOrg/content/pdf/welcome_to_the_college_board_fullversion.pdf.

College Board. 2019. *Enrollment Planning Service: Best Practices and Getting Started*. PDF. https://highered.collegeboard.org/media/pdf/eps-whitepaper.pdf.

College Board. 2021. Pricing and Program Updates Coming to Search This Fall. https://allaccess.collegeboard.org/pricing-and-program-updates-coming-search-fall.

College Board. 2022. *Enrollment Planning Service (EPS) Snapshot*. College Board; PDF. https://highered.collegeboard.org/media/pdf/eps-snapshot.pdf.

College Board. 2023a. *Connections Geographies: Detailed Connections Geography Maps*. https://drive.google.com/file/d/1d7nVl5yL6txrG84o6l7Bg5RerqC457eS/view?usp=drive_link.

College Board. 2023b. *Detailed Connections Geography Maps: Reach More Students Where They Are*. College Board. https://cbsearch.collegeboard.org/media/pdf/connections-geographies.pdf.

College Board. n.d. *SAT Trends Dashboard Report: Interpretive Guide*. College Board. https://satsuite.collegeboard.org/media/pdf/sat-trends-dashboard-interpretive-guide.pdf.

Collins, Randall. 1994. *Four Sociological Traditions*. Oxford University Press.

Cottom, Tressie McMillan. 2017. *Lower Ed: The Troubling Rise of for-Profit Colleges in the*

New Economy. The New Press.

Domina, Thurston, Andrew Penner, and Emily Penner. 2017. "Categorical Inequality: Schools as Sorting Machines." *Annual Review of Sociology* 43: 311–30. https://doi.org/10.1146/annurev-soc-060116-053354.

Doyle, William R. 2010. "Changes in Institutional Aid, 1992-2003: The Evolving Role of Merit Aid." *Research in Higher Education* 51 (8): 789–810. https://doi.org/10.1007/s11162-010-9177-0.

Espeland, Wendy N., and Mitchell L. Stevens. 1998. "Commensuration as a Social Process." *Annual Review of Sociology* 24: 313–43. https://doi.org/10.1146/annurev.soc.24.1.313.

Federal Trade Commission. 2016. "Follow the Lead" Workshop. http://www.ftc.gov/system/files/documents/reports/staff-perspective-follow-lead/staff_perspective_follow_the_lead_workshop.pdf.

Fourcade, M. 2007. "Theories of Markets and Theories of Society." *American Behavioral Scientist* 50 (8): 1015–34. https://doi.org/10.1177/0002764207299351.

Fourcade, Marion. 2011. "Cents and Sensibility: Economic Valuation and the Nature of "Nature"." *American Journal of Sociology* 116 (6): 1721–77. https://doi.org/10.1086/659640.

Goodman, Joshua, Oded Gurantz, and Jonathan Smith. 2020. "Take Two! SAT Retaking and College Enrollment Gaps." *American Economic Journal: Economic Policy* 12 (2): 115–58.

Granovetter, Mark S. 1985. "Economic Action and Social Structure: The Problem of Embeddedness." Journal Article. *American Journal of Sociology* 91 (3): 481–510.

Hamilton, Laura T., Heather Daniels, Christian Michael Smith, and Charlie Eaton. 2024. "The for-Profit Side of Public u: University Contracts with Online Program Managers." *Socius* 10: 23780231231214952. https://doi.org/10.1177/23780231231214952.

Hirschman, Daniel, and Emily Adlin Bosk. 2020. "Standardizing Biases: Selection Devices and the Quantification of Race." *Sociology of Race and Ethnicity* 6 (3): 348–64. https://doi.org/10.1177/2332649219844797.

Hirschman, Daniel, and Laura Garbes. 2019. "Toward an Economic Sociology of Race." *Socio-Economic Review* 19 (3): 1171–99. https://doi.org/10.1093/ser/mwz054.

Holland, Megan M. 2019. *Divergent Paths to College: Race, Class, and Inequality in High Schools*. Rutgers University Press. https://doi.org/10.36019/9780813590288.

Howell, Jessica, Michael H. Hurwitz, Zachary Mabel, and Jonathan Smith. 2021. *Participation in Student Search Service Is Associated with Higher College Enrollment and Completion*. College Board. https://cbsearch.collegeboard.org/pdf/college-outreach-and-student-outcomes.pdf.

Hsu, Greta, and Amanda Sharkey. 2025. "Same Policy, No Standardized Outcome: How Admissions Values and Institutional Priorities Shape the Effect of Test-Optional Policies on Campus Diversity." *American Sociological Review* 90 (5): 813–47. https://doi.org/10.1177/00031224251352432.

Huang, Tiffany J. 2023. "Translating Authentic Selves into Authentic Applications: Private College Consulting and Selective College Admissions." *Sociology of Education*, ahead of print. https://doi.org/10.1177/00380407231202975.

Karabel, Jerome. 2005. *The Chosen: The Hidden History of Admission and Exclusion at Harvard, Yale, and Princeton*. Houghton Mifflin Co.

Khan, Shamus Rahman. 2010. "Getting in: How Elite Schools Play the College Game." In *Educating Elites Class Privilege and Educational Advantage*, edited by Adam Howard and Rubén A. Gaztambide-Fernandez. Rowman & Littlefield. https://ebookcentral.proquest.com/lib/umichigan/detail.action?docID=634205.

Khan, Shamus Rahman. 2011. *Privilege: The Making of an Adolescent Elite at St. Paul's School*. Princeton Studies in Cultural Sociology. Princeton University Press.

Koppenheffer, Michael. 2021. Introducing Enroll360: Unlock the Power of a Thriving Recruitment Ecosystem. EAB. https://eab.com/insights/blogs/enrollment/introducing-enroll360-recruitment-ecosystem/.

Korver-Glenn, Elizabeth. 2022. *Race Brokers: Housing Markets and Racial Segregation in 21st century Urban America*. Oxford University Press.

Krippner, Greta R. 2024. "Gendered Market Devices: The Persistence of Gender Discrimination in Insurance Markets." *American Journal of Sociology* 130 (3): 595–643. https://doi.org/10.1086/732318.

Lake Forest College. 2026. *Admissions Staff at Lake Forest*. College website. https:

//www.lakeforest.edu/admissions/connect-with-your-admissions-counselor.

Lee, Elizabeth M, and Rory Kramer. 2013. "Out with the Old, in with the New? Habitus and Social Mobility at Selective Colleges." *Sociology of Education* 86 (1): 18–35.

Litten, Larry H., Daniel J. Sullivan, and David L. Brodigan. 1983. *Applying Market Research in College Admissions*. College Entrance Examination Board.

MacKenzie, Donald, and Yuval Millo. 2003. "Constructing a Market, Performing Theory: The Historical Sociology of a Financial Derivatives Exchange." *American Journal of Sociology* 109 (1): 107–45. https://doi.org/10.1086/374404.

McDonough, Patricia M. 1997. *Choosing Colleges: How Social Class and Schools Structure Opportunity*. State University of New York Press.

McPherson, Michael S., and Morton Owen Schapiro. 1998. *The Student Aid Game*. Princeton University Press.

Montecalvo, Brianna. 2026. *Admissions Experience at the University of Rhode Island*. LinkedIn profile. https://www.linkedin.com/in/brianna-montecalvo-18b237b0/.

Müller-Seitz, Gordon, and Jörg Sydow. 2012. "Maneuvering Between Networks to Lead – a Longitudinal Case Study in the Semiconductor Industry." *Long Range Planning* 45 (2–3): 105–35. https://doi.org/10.1016/j.lrp.2012.02.001.

Muniesa, F., Y. Millo, and M. Callon. 2007. "An Introduction to Market Devices." *Sociological Review* 55: 1–12. https://doi.org/10.1111/j.1467-954X.2007.00727.x.

Noel-Levitz. 1998. *National Enrollment Management Survey: Findings for Fall 1997 Four-Year Institutions*. Noel-Levitz.

Norris, Davon. 2021. "Embedding Racism: City Government Credit Ratings and the Institutionalization of Race in Markets." *Social Problems*, ahead of print. https://doi.org/10.1093/socpro/spab066.

Oracle. n.d. *Loading and Assigning EPS Market Codes*. Oracle. https://docs.oracle.com/cd/E29376_01/hrcs90r5/eng/psbooks/lsad/chapter.htm?File=lsad/htm/lsad40.htm.

Pinzur, D., and T. Duterme. 2025. "Market Devices and Infrastructures: How They Differ and Why It Matters." Journal Article. *Journal of Cultural Economy* 18 (2): 212–29. https://doi.org/10.1080/17530350.2024.2409798.

Pollock, Neil, and Luciana D'Adderio. 2012. "Give Me a Two-by-Two Matrix and I Will Create the Market: Rankings, Graphic Visualisations and Sociomateriality." *Accounting, Organizations and Society* 37 (8): 565–86. https://doi.org/10.1016/j.aos.2012.06.004.

Porter, Corin. 2026. *Admissions Experience at UMass Boston and Binghamton University*. LinkedIn profile. https://www.linkedin.com/in/corin-porter/details/experience/.

Quillian, Lincoln, and Carson Phillips. 2023. *Digital Redlining in College Recruitment: How Exposure to College Advertisements Differs Across Geographic Locations*. https://www.asanet.org/wp-content/uploads/2023/06/ASA-2023_FINAL.pdf.

Reardon, Sean F., and Kendra Bischoff. 2011. "Income Inequality and Income Segregation." *American Journal of Sociology* 116 (4): 1092–153. https://doi.org/10.1086/657114.

Rodriguez, A., and K. M. McGuire. 2019. "More Classes, More Access? Understanding the Effects of Course Offerings on Black-White Gaps in Advanced Placement Course-Taking." *Review of Higher Education* 42 (2): 641–79. https://doi.org/10.1353/rhe.2019.0010.

Ruffalo Noel-Levitz. 2022a. *2022 Cost of Recruiting an Undergraduate Student Report*. Ruffalo Noel-Levitz. https://www.ruffalonl.com/thank-you/download-the-2022-cost-of-recruiting-an-undergraduate-student-report/.

Ruffalo Noel-Levitz. 2022b. *2022 Marketing and Recruitment Practices for Undergraduate Students Report*. Ruffalo Noel-Levitz. https://www.ruffalonl.com/papers-research-higher-education-fundraising/marketing-and-recruitment-practices-for-undergraduate-students/.

Saint Mary's College. 2024. *Meet the Admission Staff*. College website. https://www.saintmarys.edu/admission-aid/first-year-students/connect-with-saint-marys/admission.

Sewell, William H., and Vimal P. Shah. 1967. "Socioeconomic Status, Intelligence, and Attainment of Higher Education." *Sociology of Education* 40 (1): 1–23. <Go to ISI>://A1967ZD96500001.

Sewell, William H., and Vimal P. Shah. 1968. "Parents' Education and Children's Educational Aspirations and Achievements." *American Sociological Review* 33 (2): 191–209. <Go to ISI>://A1968ZA11500001.

Stevens, Mitchell L. 2007. *Creating a Class: College Admissions and the Education of Elites*. Harvard University Press.

Stevens, Mitchell L., Elizabeth A. Armstrong, and Richard Arum. 2008. "Sieve, Incubator,

Temple, Hub: Empirical and Theoretical Advances in the Sociology of Higher Education." *Annual Review of Sociology* 34: 127–51. https://doi.org/10.1146/annurev.soc.34.040507.134737.

Stevens, Mitchell L., and Ben Gebre-Medhin. 2016. "Association, Service, Market: Higher Education in American Political Development." *Annual Review of Sociology* 42: 121–42. https://doi.org/10.1146/annurev-soc-081715-074240.

Takamiya, Toshiro. 2005. *Mechanisms for Marketing Higher Education Information Services: The Case of the College Board*. University of Pennsylvania.

Technolutions. 2026a. *Getting Started with Staff Assignments*. Slate Knowledge Base. https://knowledge.technolutions.net/docs/getting-started-with-staff-assignments.

Technolutions. 2026b. *Querying for College Fairs*. Slate Knowledge Base. https://knowledge.technolutions.net/docs/querying-for-college-fairs.

Technolutions. 2026c. *Querying for Territory Activity*. Slate Knowledge Base. https://knowledge.technolutions.net/docs/querying-for-territory-activity.

Technolutions. 2026d. *Rule Efficiency*. Slate Knowledge Base. https://knowledge.technolutions.net/v1/docs/rule-efficiency.

Technolutions. n.d. *Slate*. Product website. Accessed October 19, 2026. https://technolutions.com/solutions/slate.

University of Chicago. n.d. *UChicago Admissions Office*. University of Chicago. https:

//collegeadmissions.uchicago.edu/contact/.

Warren, Kai. 2025. *Admissions Counselor Experience at the University of Southern Indiana.* LinkedIn profile. https://www.linkedin.com/in/kaiwarren/details/experience/.

Zemsky, Robert, and Penney Oedel. 1983. *The Structure of College Choice.* College Entrance Examination Board.

Figures

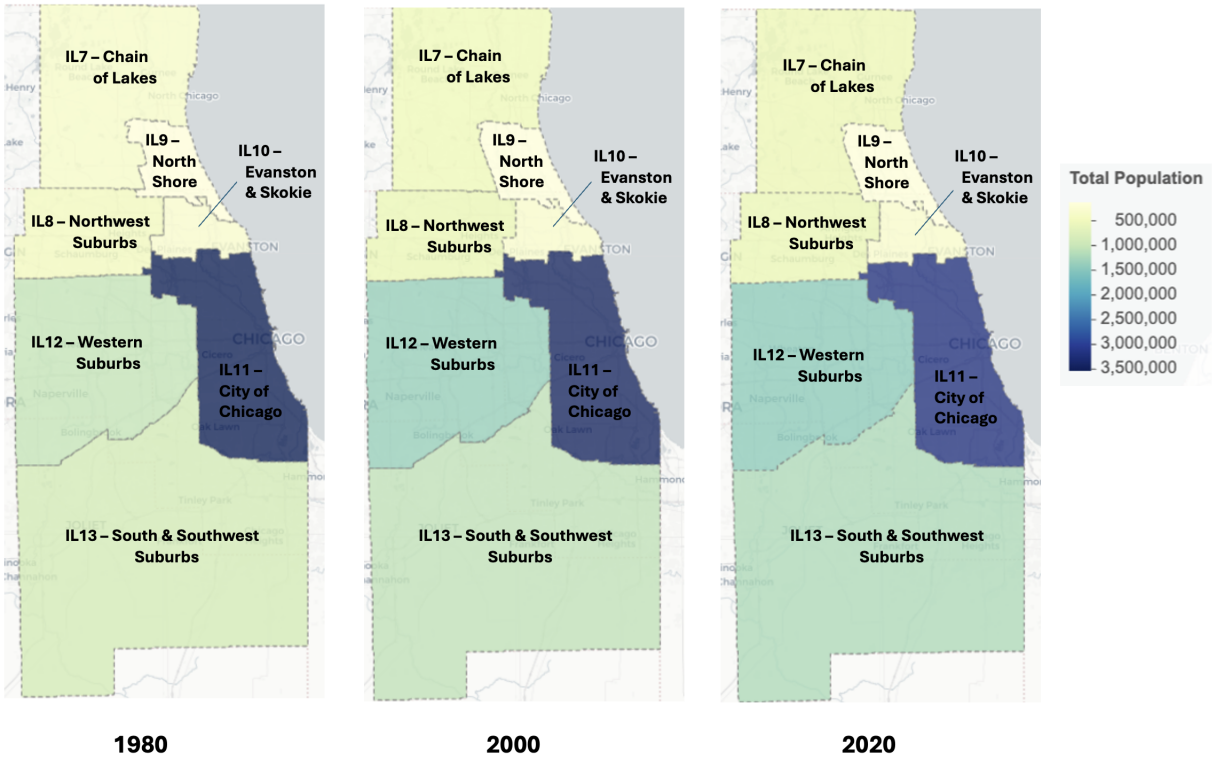


Figure 1: Population of Chicago area Geomarkets over time

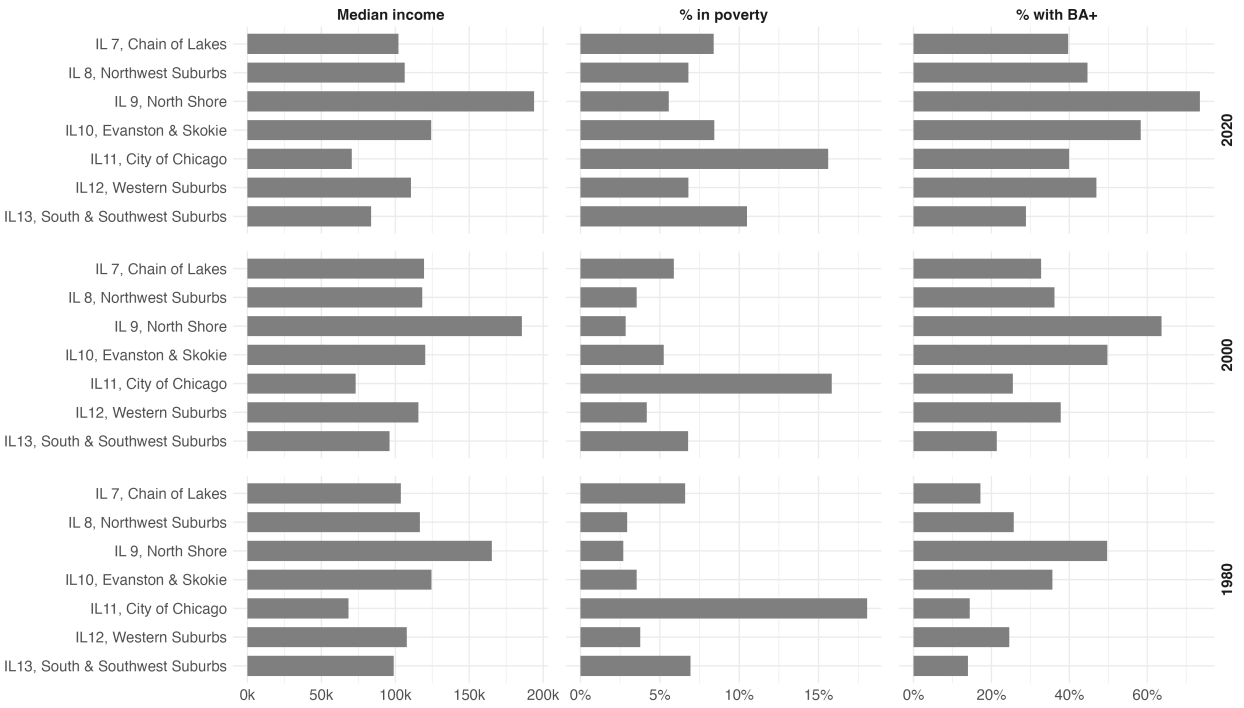


Figure 2: Socioeconomic characteristics of Chicago area Geomarkets

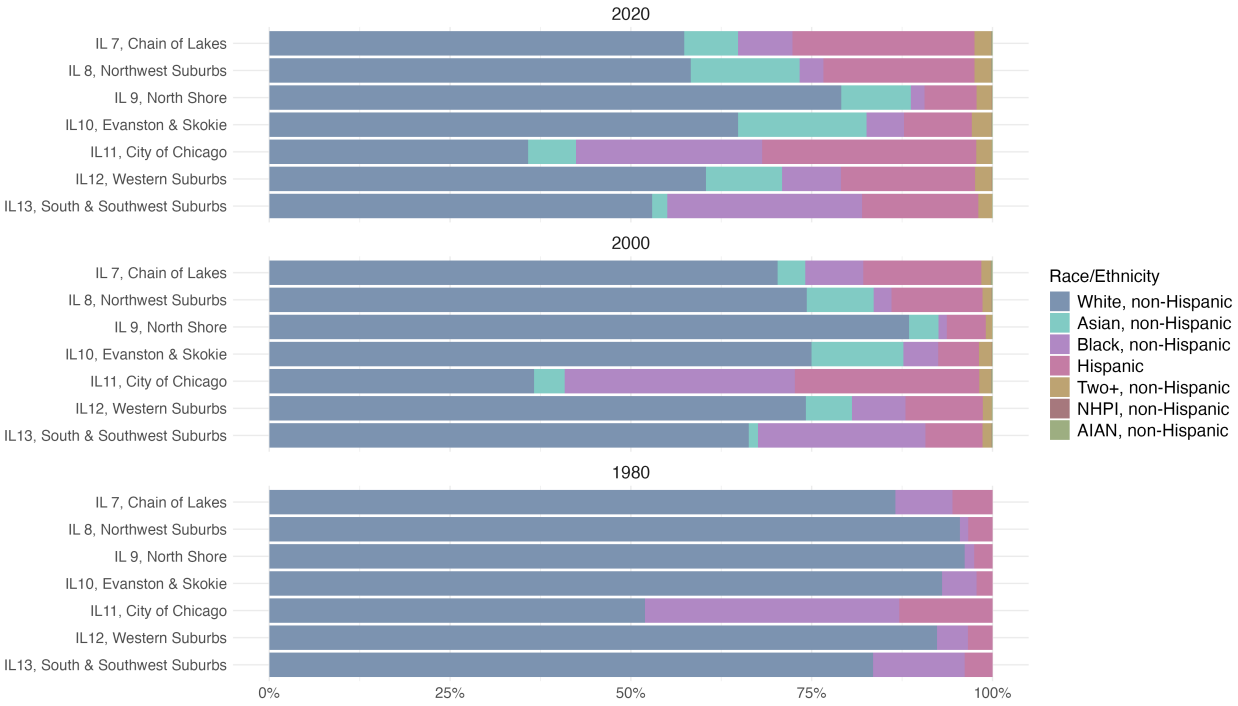


Figure 3: Racial/ethnic composition of Chicago area Geomarkets

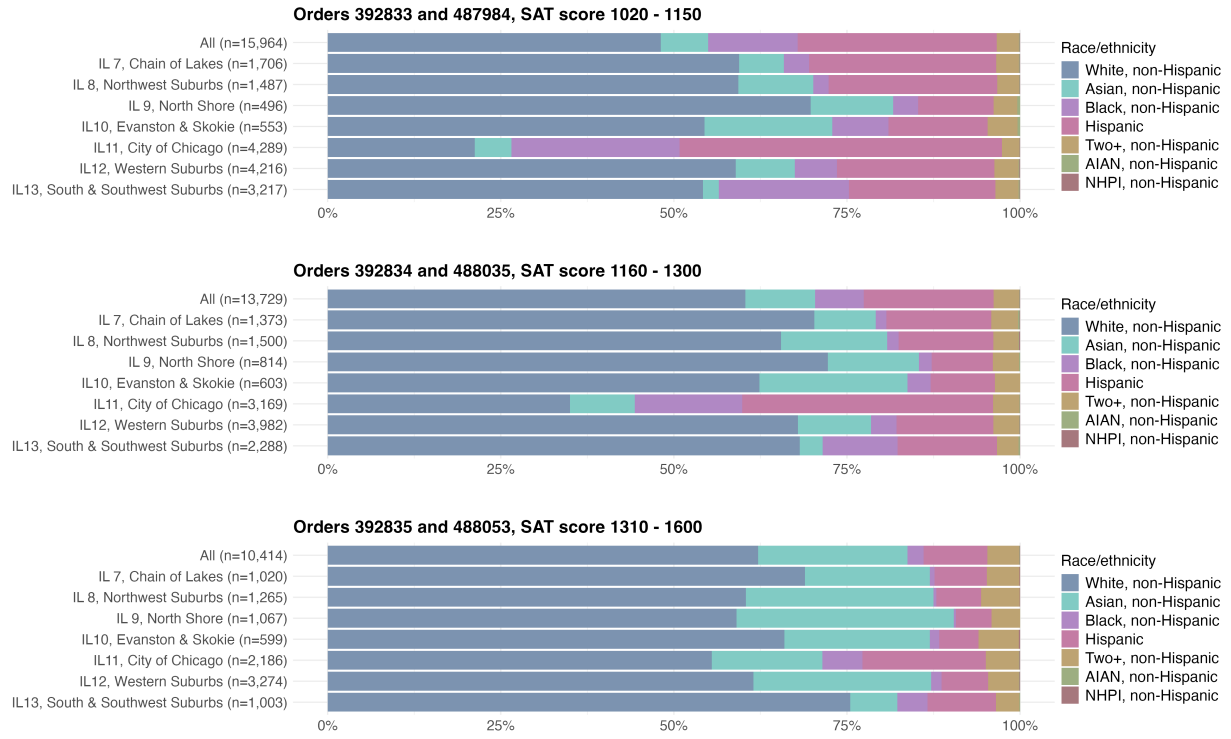


Figure 4: Racial/ethnic composition of purchased student profiles by Geomarket, Chicago area

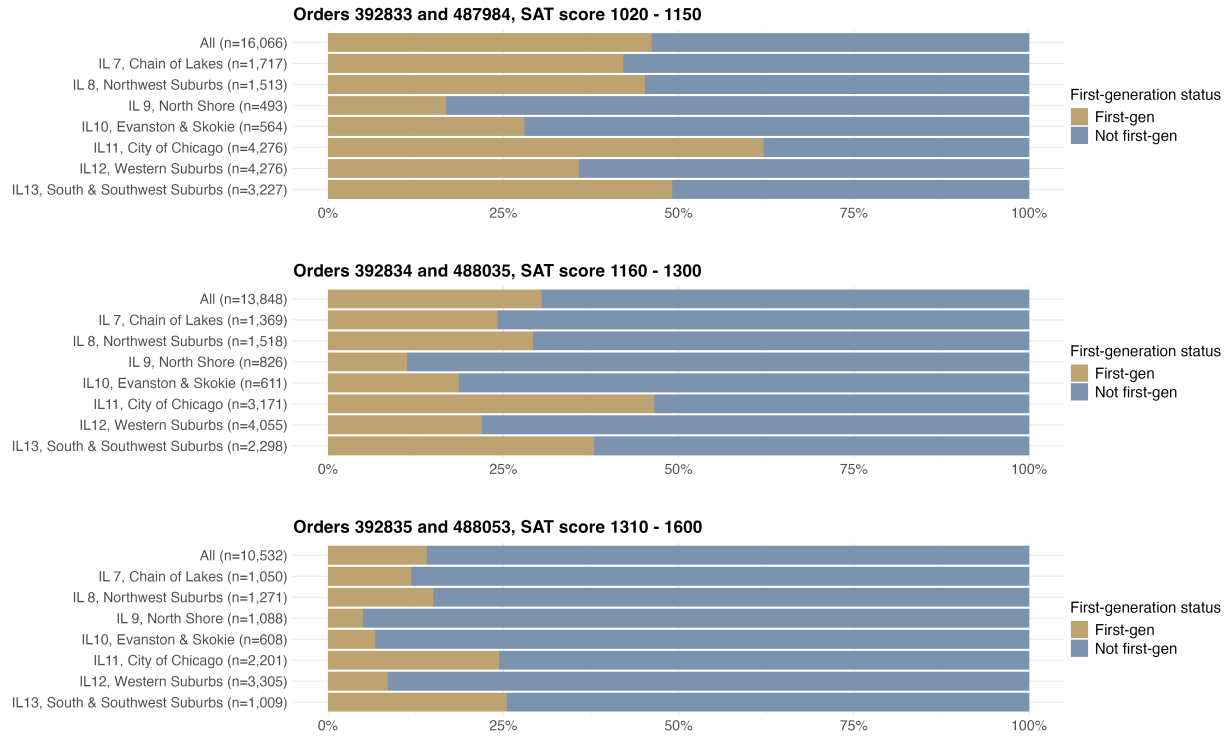


Figure 5: First-generation status composition of purchased student profiles by Geomarket, Chicago area

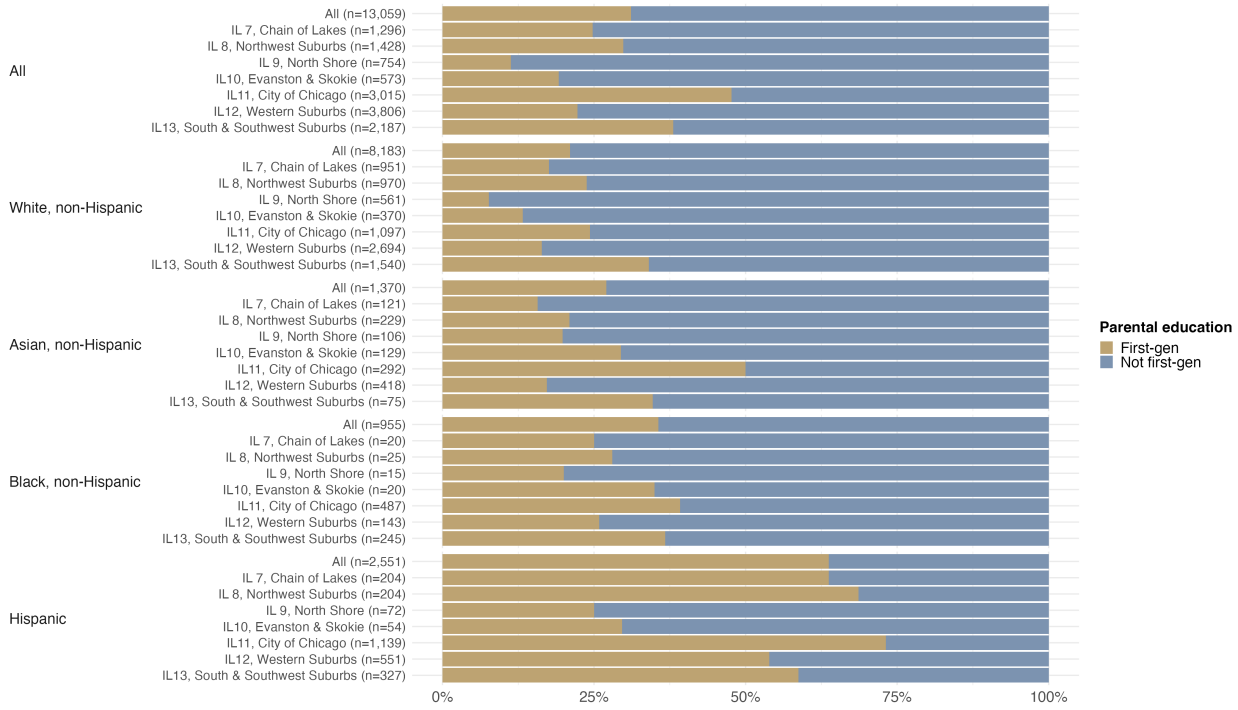


Figure 6: First-generation status by race for Chicago area Geomarkets, SAT score 1160 - 1300

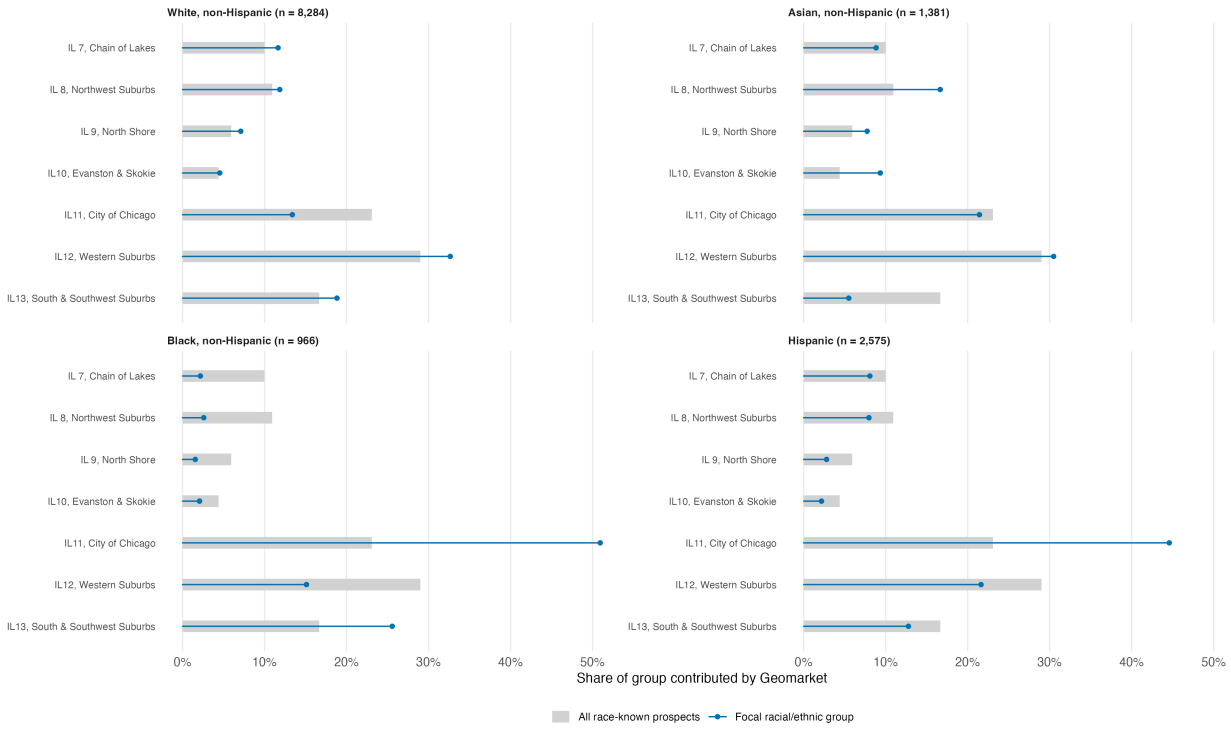


Figure 7: Geomarket contributions to each racial/ethnic group: Chicago, SAT score 1160–1300

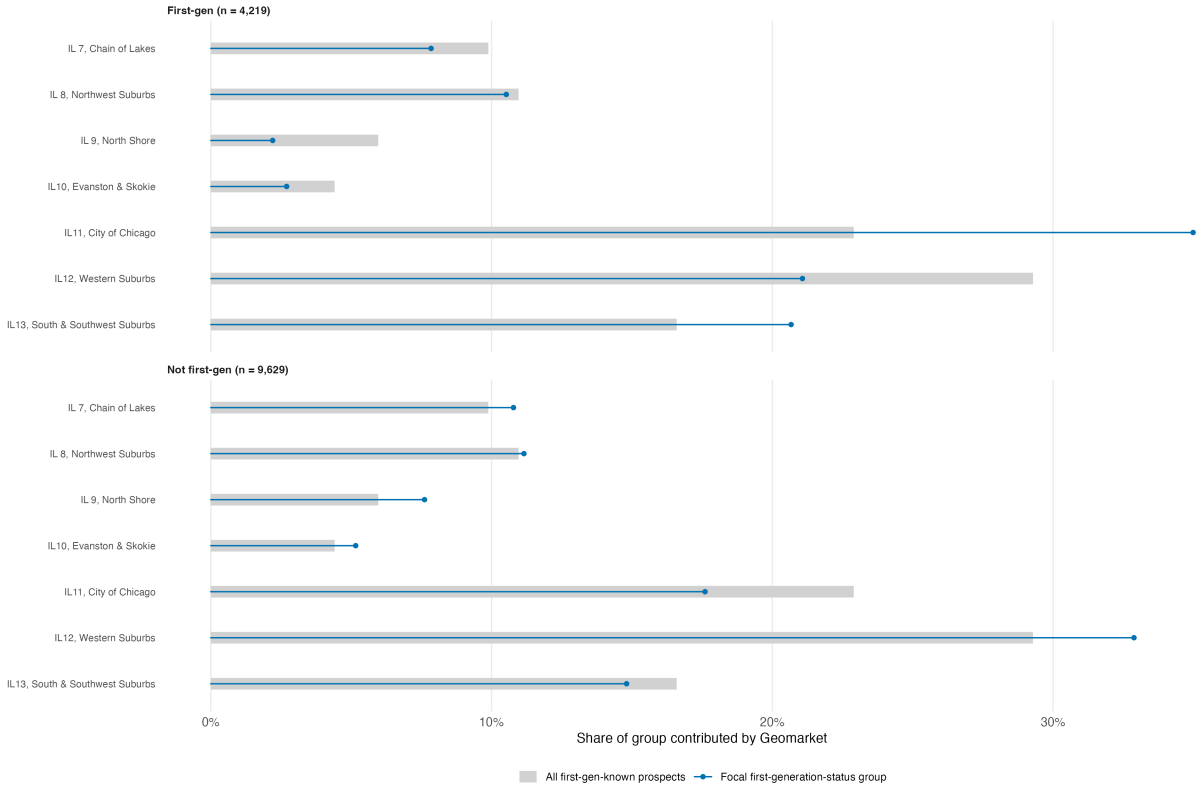


Figure 8: Geomarket contributions by first-generation status: Chicago, SAT score 1160–1300

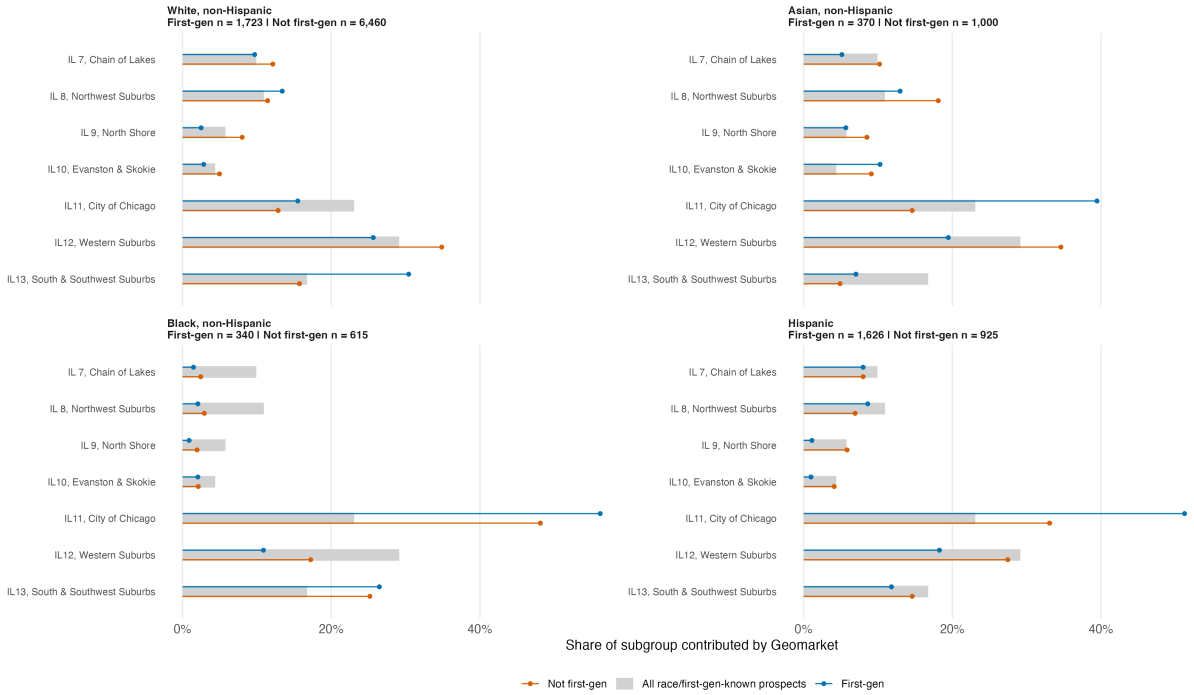


Figure 9: Geomarket contributions by race and first-generation status: Chicago, SAT score 1160–1300

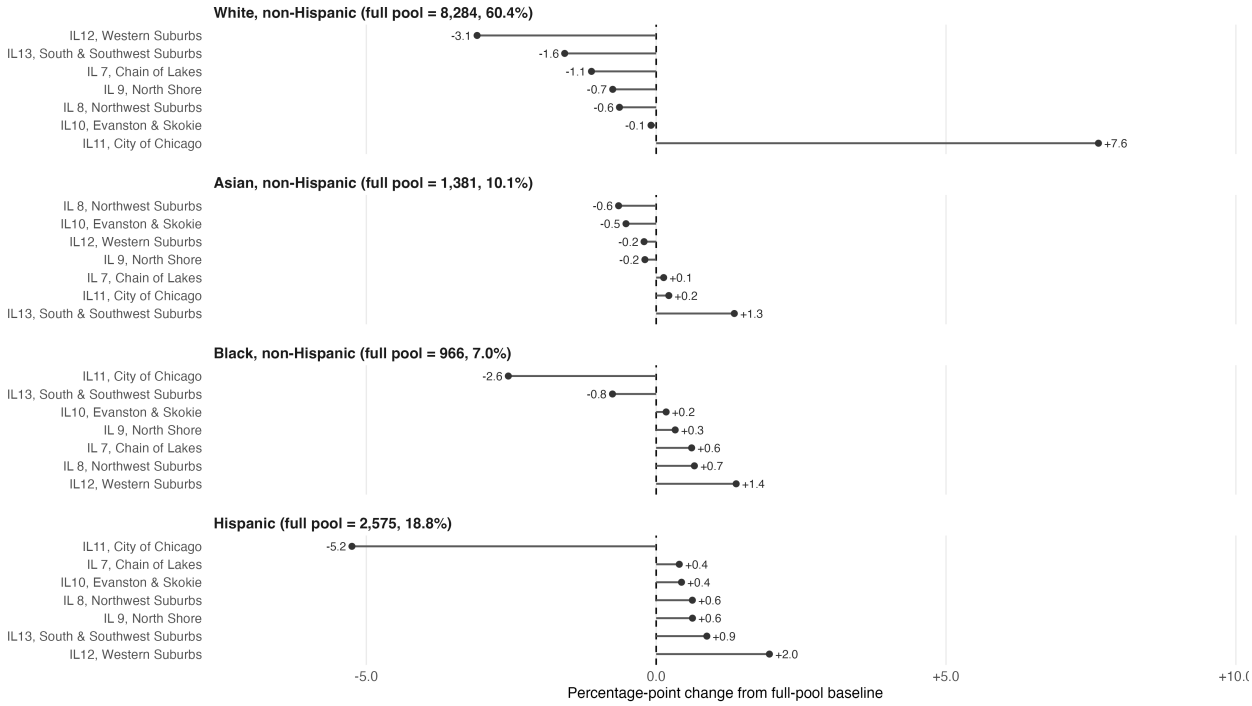


Figure 10: Percentage-point change in racial/ethnic composition after excluding each Geomarket: Chicago, SAT score 1160–1300

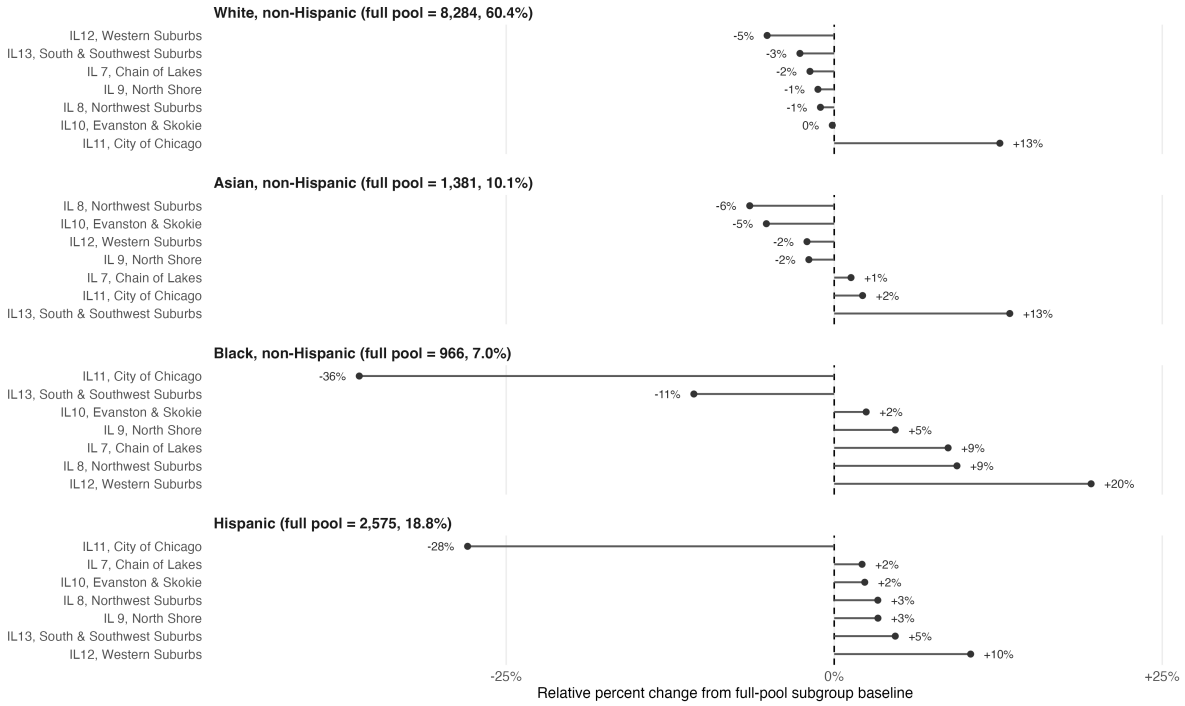


Figure 11: Relative percent change in racial/ethnic composition after excluding each Geomarket: Chicago, SAT score 1160–1300

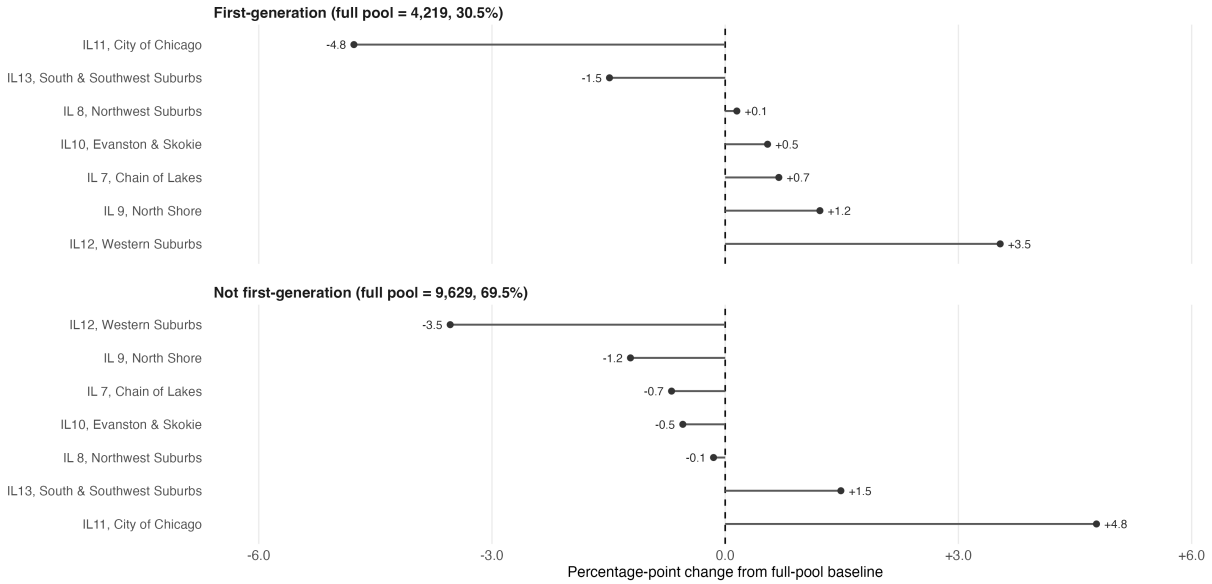


Figure 12: Percentage-point change in first-generation status composition after excluding Geomarket: Chicago, SAT score 1160–1300

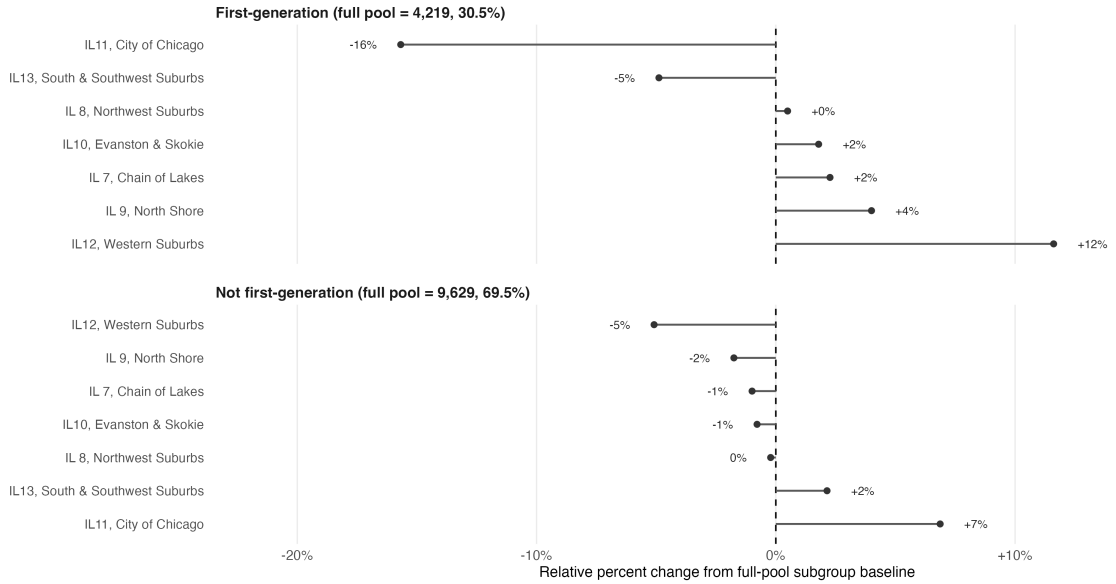


Figure 13: Relative percent change in first-generation status composition after excluding Geomarket: Chicago, SAT score 1160–1300

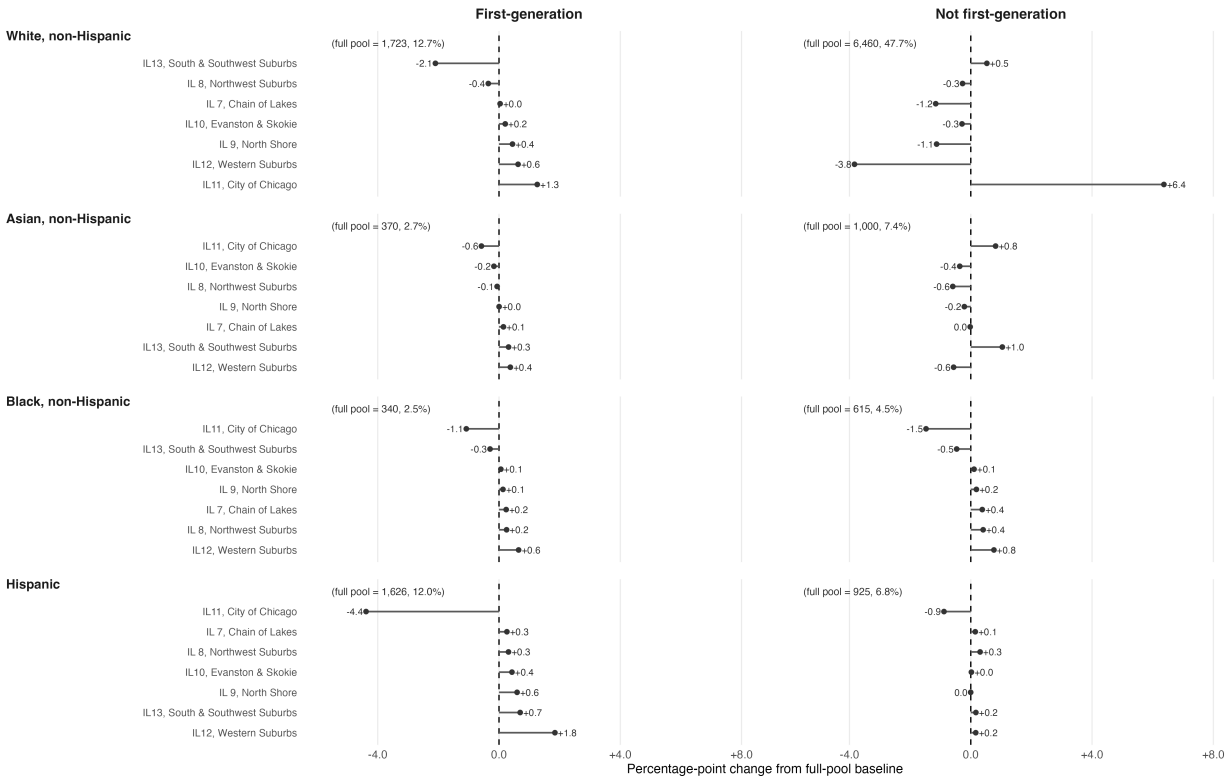


Figure 14: Percentage-point change in race × first-generation composition after excluding Geomarket: Chicago, SAT score 1160–1300

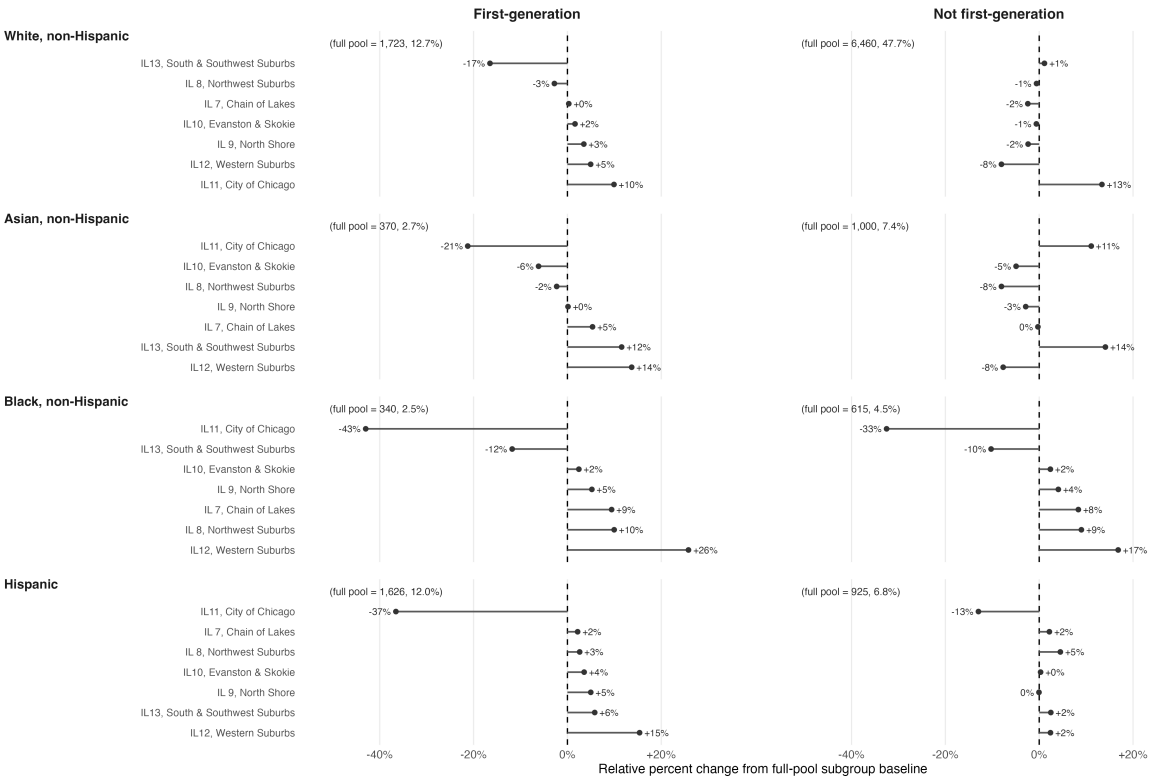


Figure 15: Relative percent change in race × first-generation composition after excluding Geomarket: Chicago, SAT score 1160–1300